HEC Paris
Master in Management, Business Analytics Using Python

Which Citi Bike stations will run out of bikes tomorrow morning?

A predictive rebalancing model for New York City's bike-share network

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Written report, Group 01

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October 2026

Data: Citi Bike System Data (50.8 million rides), station status feed (1,063 million snapshots), Meteostat weather, Apple App Store reviews. All analysis in Python (pandas, scikit-learn, XGBoost, imbalanced-learn, mlxtend, gensim).

Executive summary

Citi Bike moves bikes by truck, mostly at night, so the list of stations to refill must exist *the evening before*. The question is therefore: *which stations will be empty tomorrow morning, and should the trucks give priority to commuter or leisure stations?*

We combined four public sources (50.8 million trips, 1,063 million two-minute station snapshots, daily weather and 500 App Store reviews) into 851,613 station-days covering 2,318 stations. We defined a shortage as a station empty for at least one hour of the 7–10am window, which happens on only 4.7% of station-days. The main findings are:

1. **Shortages can be predicted the evening before.** Using only information known at night, XGBoost reaches an AUC of 0.906 on a held-out set of 30,000 station-days (0.899 in cross-validation), ahead of a random forest (0.893), a logistic regression (0.865) and a decision tree (0.842).
2. **The model beats the obvious rule.** Against the practical rule "visit the stations that were empty longest today", a list of 10% of the stations catches 62% of the next morning's shortages instead of 51%, and wins on 68 of the 77 unseen test nights.
3. **Commuter stations come first.** Residential stations make up 28% of station-days but 43% of the drop-off list. Work stations behave in the opposite way and fill up in the morning.
4. **Trucks can also collect.** Among stations that are full tonight and ranked at low risk, 0.1% run empty the next morning and 53% are full again, so they are natural pick-up points.
5. **The model is stable.** Trained on August–October 2025 only, its AUC stays between 0.86 and 0.92 over the following year, while a demand model drops to $R^2 = 0.63$ in January.
6. **Riders are only partly separable.** A member versus casual classifier reaches an AUC of 0.72 at best, but the 28% of casual rides made at weekday commuting peaks (40% for members) define a target segment for membership offers.

Our answer: *each evening, rank all stations by the probability of being empty tomorrow, refill residential stations first with bikes taken from stations that are full and at low risk, and treat chronic stations structurally rather than with a daily truck.*

How this report is organised

The report follows one thread, from a customer complaint to an operating rule.

Section 1 states the operational problem and the five hypotheses we tested. **Section 2** describes the data: four sources and the cleaning rules, then a first look at the rhythm of the network. **Section 3** is the analysis. It first *understands* the network (station types, drivers of demand, who rides). It then *predicts* tomorrow's empty stations and checks that the model survives a change of season. **Section 4** turns the results into recommendations and a decision rule for the length of the nightly list. **Section 5** compares results with hypotheses and states the limits. **Section 6** concludes.

Every figure sits next to the paragraph that uses it, and each result is followed by what it changes for the operator.

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1 Introduction

This section explains why a bike-share operator cares about empty stations. It then states our question and the hypotheses we want to test.

1.1 The problem: empty stations

A bike-share system is a network in which the product, a bike available at the station where the customer stands, is constantly consumed by the customers' own behaviour. Everybody leaves the residential areas at 8am and everybody leaves the offices at 6pm. Without intervention, residential stations empty in the morning while work stations overflow, and the reverse happens in the evening.

The cost of this imbalance is easy to state:

- **An empty station is a lost ride.** The commuter who finds no bike takes another mode of transport. Figure 1 shows that on a weekday the share of time an average station is empty more than doubles between 6 and 11am (2.9% at 6am, 6.3% at 11am), exactly when the offices fill up.
- **A full station is a lost return.** A rider who cannot dock loses time and remembers it.
- **Both are visible to customers.** In 500 written App Store reviews, one in five is mainly about docks and stations (Section 2.4).

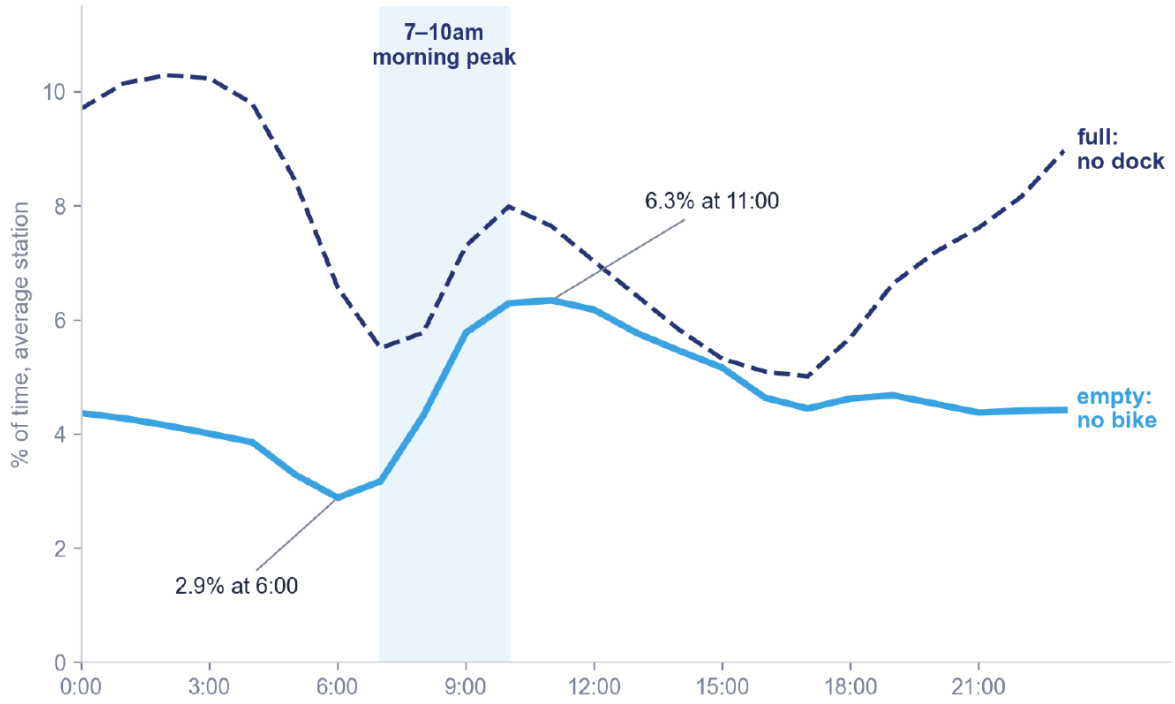


Figure 1: Share of time an average station is empty (no bike) or full (no free dock), by hour of a weekday. Average over 2,318 stations, August 2025 to August 2026, from 2-minute station snapshots.

Operators answer with rebalancing: trucks move bikes from full stations to empty ones, mostly at night. The route has to be planned the evening before, which turns a descriptive problem into a *prediction* problem with a hard information constraint: only what is known at night can be used. A second, commercial question comes with it. Casual riders make 18% of rides, and knowing which of them ride like commuters tells Citi Bike whom to offer a membership.

1.2 Research question and hypotheses

Our guiding question is: *which stations will run out of bikes tomorrow morning, and should the trucks give priority to commuter or leisure stations?* It has two parts. The first is a ranking problem: for each station and each evening, how likely is it that the station will be empty during the next morning peak? The second is a question of priority: commuter stations lose their bikes when people leave residential areas for work, whereas leisure use is spread over the weekend and the middle of the day, so the two kinds of station do not call for the same effort at the same time.

Three conditions frame the answer. The ranking may use only what is known the evening before. It is judged on days that were not used to build it. And it has to do better than a rule an operator can apply without any model, which is to visit the stations that were empty longest today.

Behind the question we formulated five hypotheses before the analysis and tested each of them (Table 1). They cover the structure of the network (H3), the drivers of demand (H1, H4), the riders (H2) and the durability of the models (H5).

	Hypothesis	Why we expected it	How we test it	Verdict
H1	Time explains a station's demand better than the station	Rides follow strong seasonal and weekly rhythms, so the calendar and the weather should explain most of a station's daily demand.	Regression of daily departures with one group of variables at a time (Section 3.3).	Not at station level
H2	Trip patterns separate members and casual riders	Members mostly commute and casual riders mostly ride for leisure, so duration, distance, hour and bike type should reveal who rides.	Five classifiers compared by AUC on unseen rides (Section 3.4).	Partly
H3	Stations fall into a few behavioural types	A station's use depends on whether it sits among homes or offices, so hourly profiles should group into a small number of types.	PCA and k-means on hourly profiles, silhouette, and a check against the map (Section 3.2).	Supported
H4	Weather is a secondary but real driver	Cold and rain stop some trips, but commuters still need to travel, so weather should move the volume without deciding which stations run short.	Weather effects in the demand regression, and removal of the weather variables from the shortage model (Sections 3.3 and 3.5).	Partly
H5	Models lose accuracy in another season	A model fitted on one season learns that season's behaviour.	Training on August to October 2025 and testing on each later month (Section 3.6).	For demand only

Table 1: The five hypotheses, the reasoning behind them, the test applied and the verdict.

Section 5 returns to each hypothesis with the evidence.

2 Data

Before any model, we need to know what the data can and cannot say. This section presents the four sources, the steps that turn them into analysis tables, and a first look at how the network behaves through the year, the week and the day.

2.1 Four sources

Table 2 lists the raw data. The trips and the station status are the core; weather and reviews give context.

Source	Provider	Size and period	Main content
Trips	Citi Bike System Data (official)	50.8 M rides, 13 columns, Aug 2025 to Aug 2026	start and end time and station, coordinates, bike type, member or casual
Station status	Kaggle, E. Rosenthal (from the Citi Bike GBFS feed, CC BY 4.0)	1,063 M snapshots, 23 columns, every 2 min, Jan 2025 to Sep 2026	bikes and free docks, capacity, in service, station name, time
Weather	Meteostat, Central Park	638 days, 11 columns, Jan 2025 to Sep 2026	average temperature, rain, wind
App reviews	Apple App Store (US)	500 reviews, 7 columns, Feb 2021 to Sep 2026	stars, title, text, app version, date

Table 2: The four data sources.

What this means for Citi Bike

The station status feed is what makes the project possible. Trip data tell us what customers *did*; they cannot tell us what customers *could not do*, because a rider who finds an empty station leaves no trace in the trips. Station snapshots reveal that unmet demand. An operator already has this feed, so the model we propose needs no new data collection.

2.2 From raw data to analysis tables

Three steps turn the raw files into the tables used by the models (Figure 2).

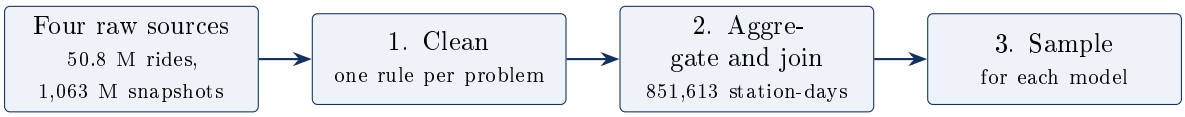


Figure 2: Data pipeline.

Step 1: cleaning. Each rule corresponds to a problem found while exploring the data (Table 3). For trips, 204,183 rides (0.4%) are removed. For station status, 643 M of 1,063 M snapshots are kept; most of those removed lie outside the study period (389.7 M). The rules are applied with pandas (`dropna`, `drop_duplicates`, boolean filters) before any aggregation.

Trips: 50,810,230 rides		Station status: 1,062,555,330 snapshots	
No start or end station	168,313	Outside the period	389.7 M
Longer than 3 hours	34,578	Station not in service	19.5 M
Duplicated ride id	512	Zero or missing capacity	8.0 M
Zero or negative duration	482	Duplicated snapshot	2.4 M
Outside Aug 2025 to Aug 2026	290	**Kept**	**642.9 M**
Under 1 minute, wrong coordinates	8	Station-days after joining	901,753
Removed in total	**204,183 (0.4%)**	Incomplete day (< 360 snapshots)	9,406
		Inactive station (< 365 departures)	40,734
		Kept	**851,613**

Table 3: Cleaning rules and rows removed.

Step 2: aggregating and joining. We aggregate to one row per station and day, with the share of the 7–10am and 4–8pm snapshots in which the station is empty or full. Rides are joined to stations by station name (the status data carry no trip identifiers, so this is a pandas `merge` on the name), and weather and calendar variables are joined by date. Stations are matched by name for 99.2% of rides. After removing incomplete days and inactive stations, the final table has **851,613 station-days** for **2,318 stations**.

Step 3: sampling. Fitting every model on 851,613 rows or 50 million trips would be slow without adding information. We sample:

- **Rides:** a 0.4% stratified sample (202,423 rides), `sample(frac=0.004, random_state=42)` by month and rider type, (26 groups). Stratifying keeps the seasonality and the 18% share of casual riders, and 0.4% still leaves at least 4,800 rides per month. Variables we did not stratify on match the full data within 0.2 points.
- **Station-days:** 100,000 for the demand model and 150,000 for the shortage models. The full table is used for benchmarks and stress tests.

2.3 First look: seasons, hours and days

The data show three regularities that shape everything that follows.

Seasons and the pull of the centre. Daily rides follow the seasons (Figure 3): about 208k rides on the busiest day (15 August 2026) against 6k on 26 January 2026. The share of casual riders follows the same cycle, from 22% in September 2025 down to 9% in January and February 2026, and back to 20–21% in summer 2026. Bikes also drift toward the centre: over a year 277 stations lose more than 300 bikes (net flow, rides ending minus rides starting) and 265 gain more than 300. The busiest 10% of stations account for 39% of rides.

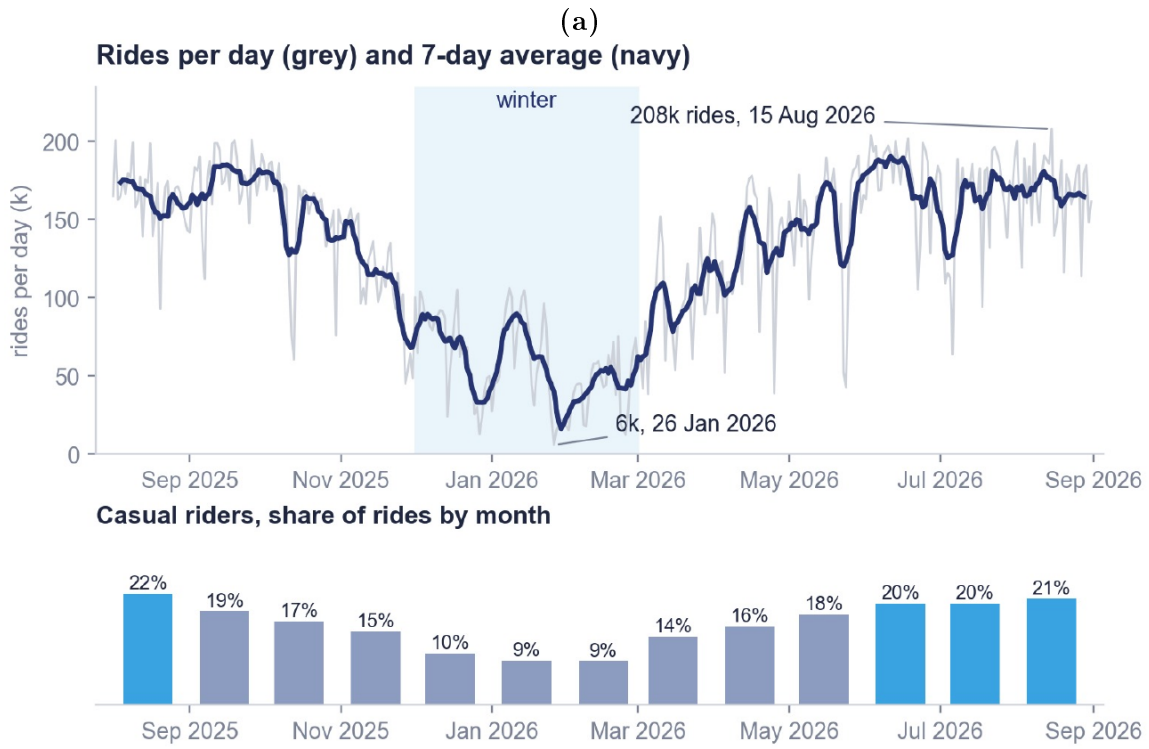


Figure 3: (a) rides per day (grey) and 7-day average (navy), and the share of casual riders by month. (b) rides started per station and net flow of bikes over the year (blue: loses more than 300 bikes; navy: gains more than 300; grey: in between).

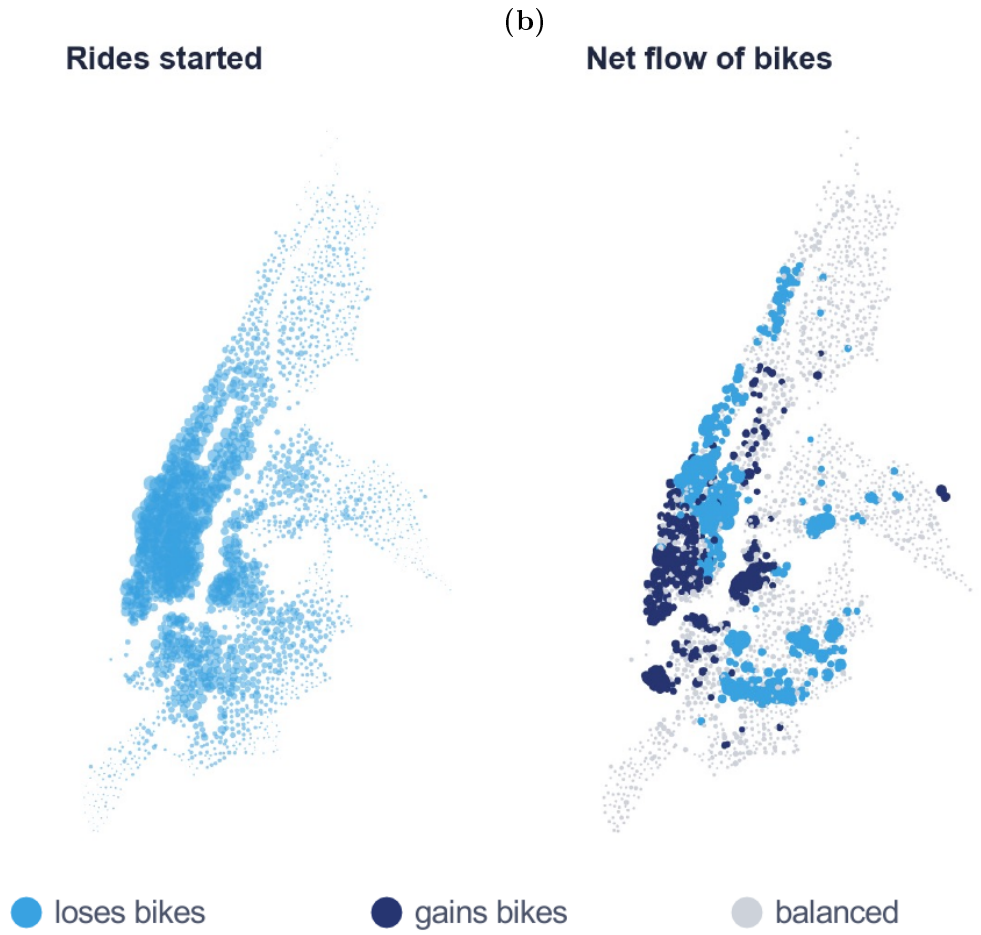


Figure 3: Panels continued.

Two peaks in the day. Weekday rides peak at 8am and between 5 and 6pm, while weekends have a single midday peak (Figure 4a). This is the signature of commuting, and it is what produces the morning shortages. Members and casual riders do not share the same weekday rhythm (Figure 4b): members have a sharp morning peak of about 8% of their weekday rides at 8am, while casual riders have a much flatter morning and a broad afternoon.

(a)

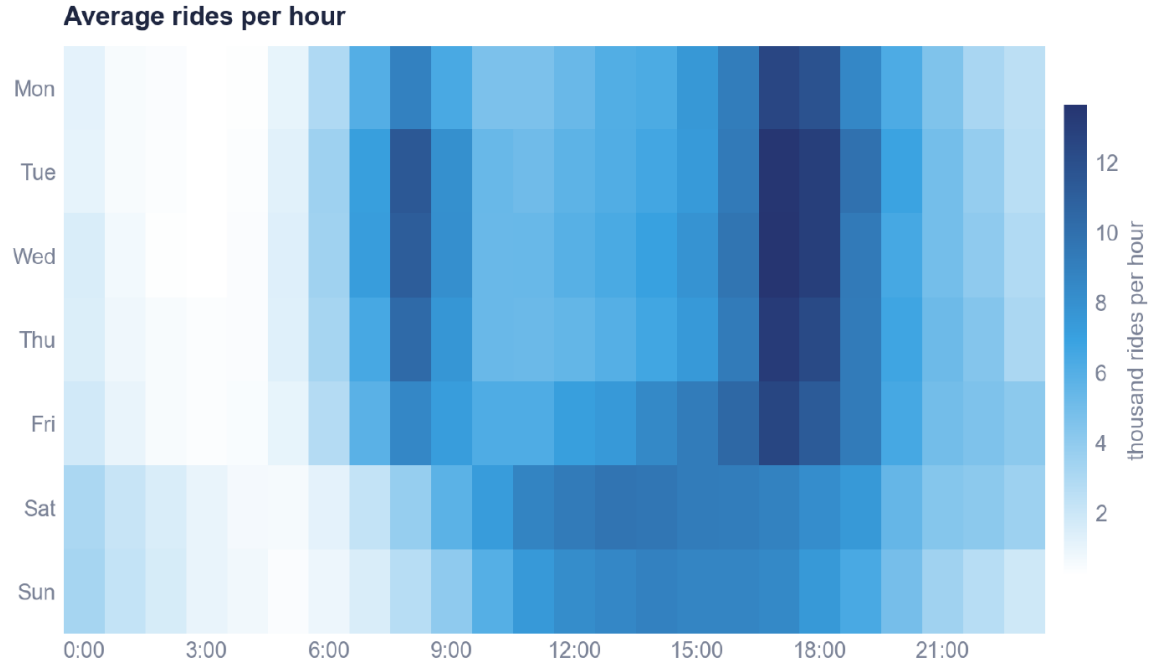


Figure 4: (a) average rides per hour by day of the week and hour. (b) share of each group's weekday rides by hour. Average rides per hour = sample rides × 250 ÷ number of such days in the year.

(b)

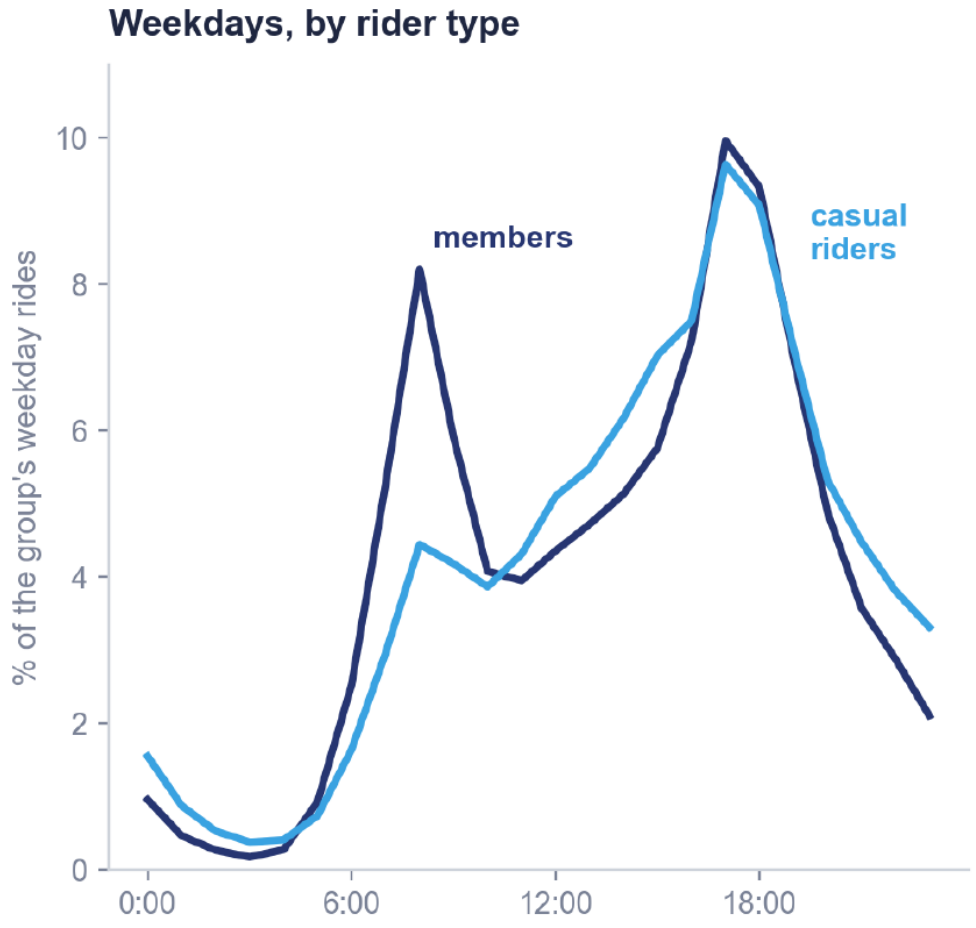


Figure 4: Panels continued.

Shortages are concentrated. Figure 5 shows that shortages are a concentrated phenomenon: the 10% of stations most often empty account for 46% of all empty time on weekday mornings. They are also seasonal and weekly: the share of station-days empty for at least an hour of 7–10am is highest in winter (7.8% in February, 6.8% in December and 6.5% in January, against about 3% in August) and on Mondays (5.1%, falling to 2.9% on Saturday). The target we later predict (empty for at least one hour of 7–10am, i.e. empty in at least 33% of the 7–10am snapshots) occurs on **4.7% of the 851,613 station-days**. A model that always answered "not empty" would be 95.3% accurate and useless. That is why accuracy is not our metric: we use the AUC and the quality of a ranked list instead.

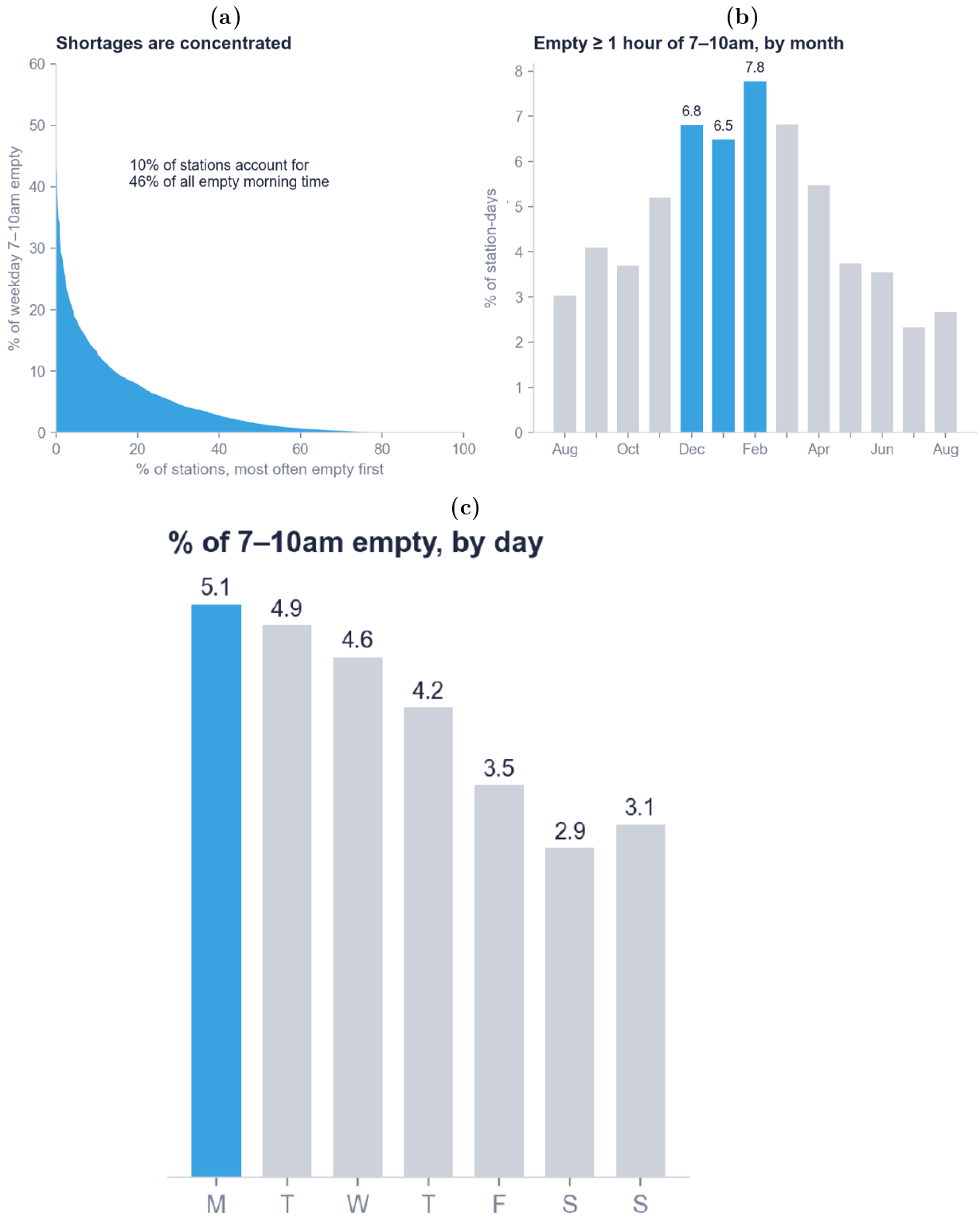


Figure 5: (a) share of weekday mornings each station is empty (stations sorted from most to least often empty). (b) share of station-days empty for at least one hour of 7–10am, by month. (c) share of 7–10am empty, by day of the week.

What this means for Citi Bike

A policy focused on a small set of stations addresses almost half of the problem, and the monthly and weekly pattern shows when to add capacity: winter and Mondays. A ranked list, rather than a yes/no flag for each station, is therefore the right output.

2.4 What riders say

The reviews are a customer-side view of the same problem. We fitted a topic model (LDA, gensim `LdaModel`) on the 500 written reviews after removing stop words and punctuation and lemmatising (so "pas" in the output comes from "pass"). The number of topics was chosen by coherence (`CoherenceModel`): four topics, with a coherence of 0.36 (Figure 6a). The main topic of a review is the topic with the highest probability.

(a)

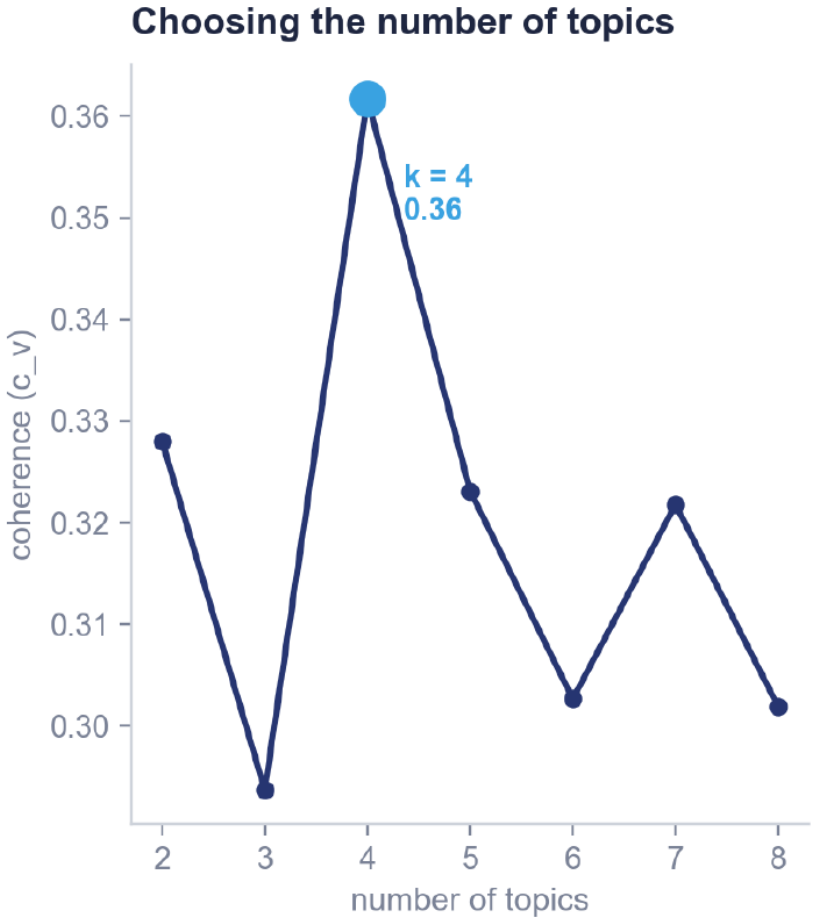


Figure 6: Topic model of the 500 reviews. (a) coherence by number of topics (four chosen). (b) the most likely words of each topic, with the number of reviews and the average rating.

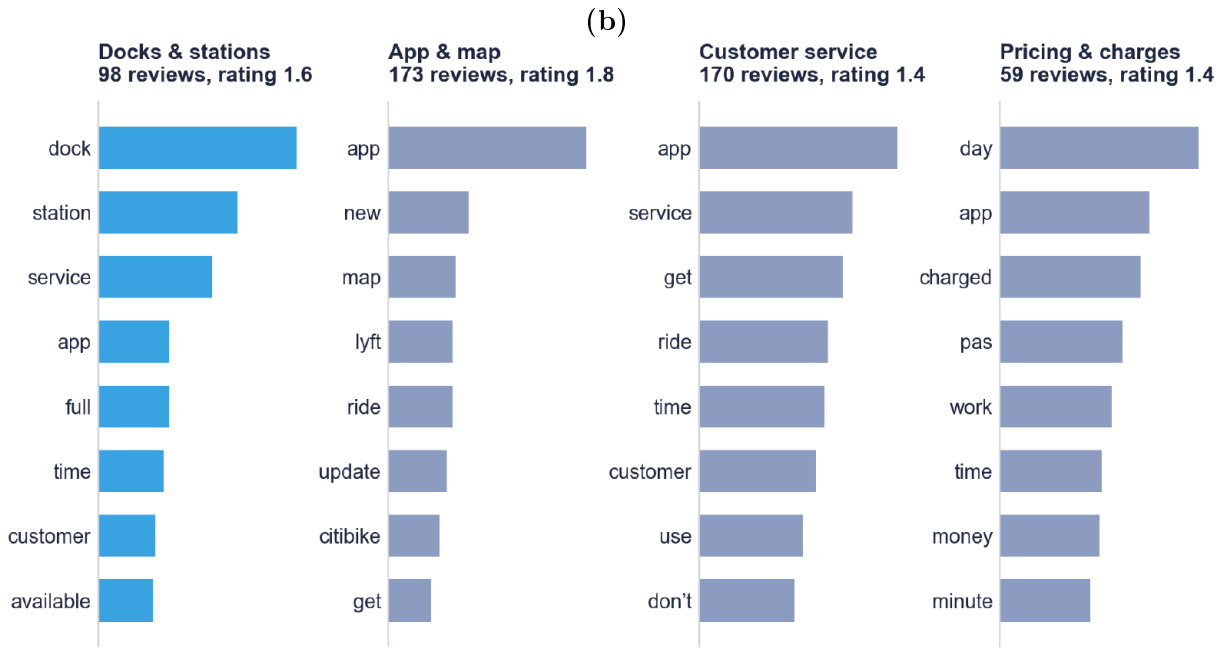


Figure 6: Panels continued.

The four topics are *docks and stations* (98 reviews, average rating 1.6), *app and map* (173 reviews, 1.8), *customer service* (170 reviews, 1.4) and *pricing and charges* (59 reviews, 1.4). Docks and stations therefore represent 20% of the reviews (Figure 7). Three examples illustrate it: "Downtown Manhattan docks are constantly full ... Defeats entire purpose" (March 2022), "It's insane how many of the bikes or stations are constantly broken, and how 'unbalanced' they are" (June 2026), and "Lots of times there are no docks available and lots of times the bikes are unusable. Still a fast way to get around" (September 2026).

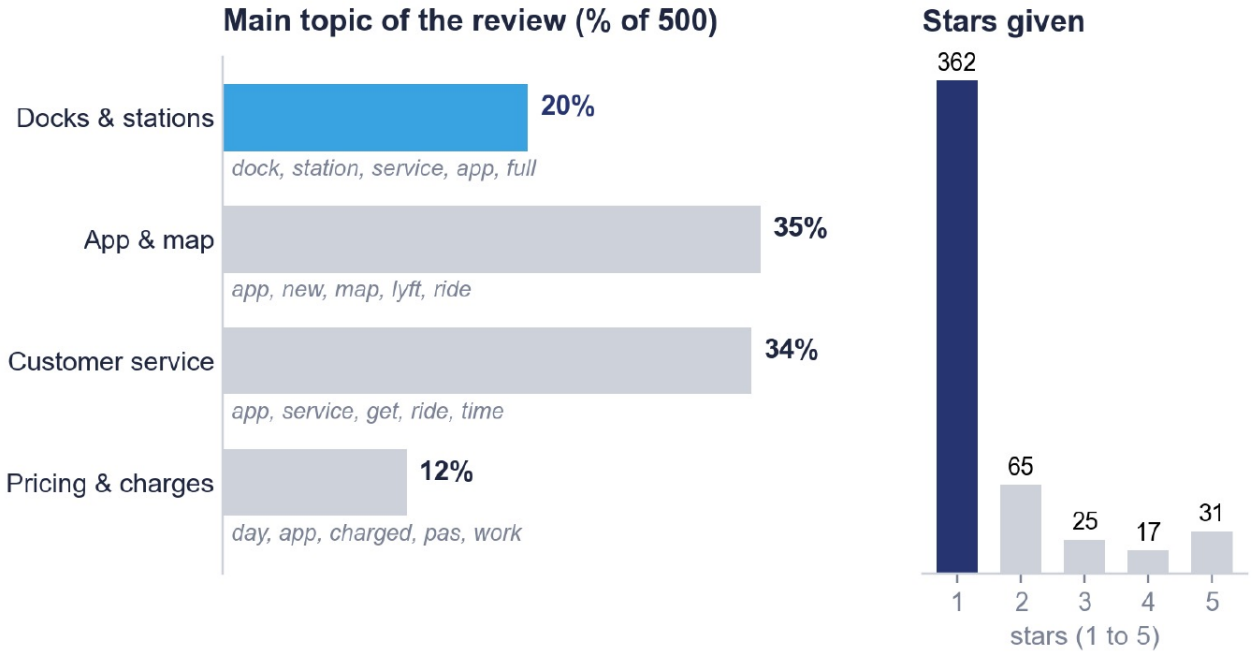


Figure 7: Main topic of the 500 reviews (percentage of reviews, with the top words of each topic) and stars given.

Caveat

Written reviews come mostly from unhappy users: 362 of 500 give one star (average 1.58), while the overall rating of the app is 4.5 out of 5 over 10,193 ratings, since most users rate without writing. We therefore use the reviews as a signal of what annoys customers, not as a satisfaction measure. The sample is small and spans more than five years.

The data show that the problem is real (Figure 1) and concentrated in a few stations, and that customers notice it. The next section moves from describing the network to understanding it.

3 Analysis and results

We first *understand* the network: which types of stations exist (3.2), what drives their demand (3.3) and who the riders are (3.4). We then *predict* tomorrow's empty stations from what is known the evening before (3.5), and finally test whether the prediction survives a change of season (3.6).

3.1 Methodological frame

We use two kinds of model. *Descriptive* models (PCA and k-means) describe how the network behaves and produce a station-type variable. *Predictive* models (regression and classifiers) are judged only on data they have not seen. For every predictive model we state the split (random or by date), the metric and the settings, so that each number can be reproduced with a fixed random seed (Table 4).

Task	Data and split	Settings
Station types	48 hourly shares per station	`StandardScaler`; `PCA(n_components=0.90)`; `KMeans(n_clusters=4, n_init=10, random_state=42)`
Demand model	100,000 station-days; 80/20 `train_test_split`, `random_state=42`	`LinearRegression` on $\log(1+\text{departures})$; Lasso on standardised variables
Member vs casual	202,423 sampled rides; 40,036 test rides	forest: 100 trees, `min_samples_leaf=20`; KNN on 20,000 training rides, standardised variables
Shortage models	150,000 station-days (`random_state=42`); 120,000 train, 30,000 test; `KFold(5, shuffle=True, random_state=10)`	tree `max_depth=5`; forest as above; XGBoost `max_depth=4`, `eta=0.3`, 100 rounds
Nightly test	all station-days, split by date: 80% of nights train, last 20% (77 nights) test	same XGBoost settings
Out-of-time test	train Aug–Oct 2025, test each later month	same settings

Table 4: Splits and settings of every model.

For classifiers, the main metric is the area under the ROC curve,

$$\text{AUC} = \mathbb{P}\big(s(X^+) > s(X^-)\big) + \tfrac{1}{2}\,\mathbb{P}\big(s(X^+) = s(X^-)\big), \tag{1}$$

the probability that a randomly chosen positive case (a station that will be empty, a casual ride) receives a higher score s than a randomly chosen negative one. It equals 0.5 for a random ranking and 1 for a perfect one. It does not depend on a threshold and holds up under class imbalance. For a given threshold we also report

$$\text{precision} = \frac{TP}{TP+FP}, \qquad \text{recall} = \frac{TP}{TP+FN}, \tag{2}$$

and, for lists of a share α of the stations, the *hit rate* (share of listed stations that really are empty) and the *share of shortages caught* (the recall of the list).

3.2 Understanding the network I: four types of stations

Stations differ a lot, so a single policy cannot fit them all. We group them by the *shape* of their day, not by their size or location.

3.2.1 Method

For each station we compute the average share of its weekday departures and of its weekday arrivals in each of the 24 hours, giving $p = 48$ variables. Using shares makes the profile independent of the station's size, so a small residential station and a large one can be compared. The pipeline has three steps:

1. **Standardise** each variable to mean 0 and variance 1 (`StandardScaler`), so that busy hours do not dominate.
2. **PCA** (`PCA(n_components=0.90)`). The principal components are the eigenvectors of the covariance matrix of the standardised data, ordered by eigenvalue λ_j; we keep the first m components such that $\sum_{j \le m} \lambda_j / \sum_{j \le p} \lambda_j \ge 0.90$. This keeps $m = 27$ components (Figure 8a), which removes noise and the strong correlation between neighbouring hours.
3. **k-means** on the 27 components. It minimises the within-cluster sum of squares

$$\sum_{c=1}^{k} \sum_{i \in C_c} \|z_i - \mu_c\|^2,$$

   and `n_init=10` restarts the algorithm ten times from different centres to avoid a poor local minimum.

3.2.2 Choosing k

The silhouette of a station i is $s_i = (b_i - a_i)/\max(a_i, b_i)$, where a_i is its mean distance to the other members of its own cluster and b_i its mean distance to the nearest other cluster. The average of s_i summarises the clustering, and values close to 1 mean well-separated groups. Ours are low and fall with k: 0.15 for $k = 2$, 0.13, 0.11 for $k = 4$ and 0.09. The elbow curve has no clear break either (Figure 8).

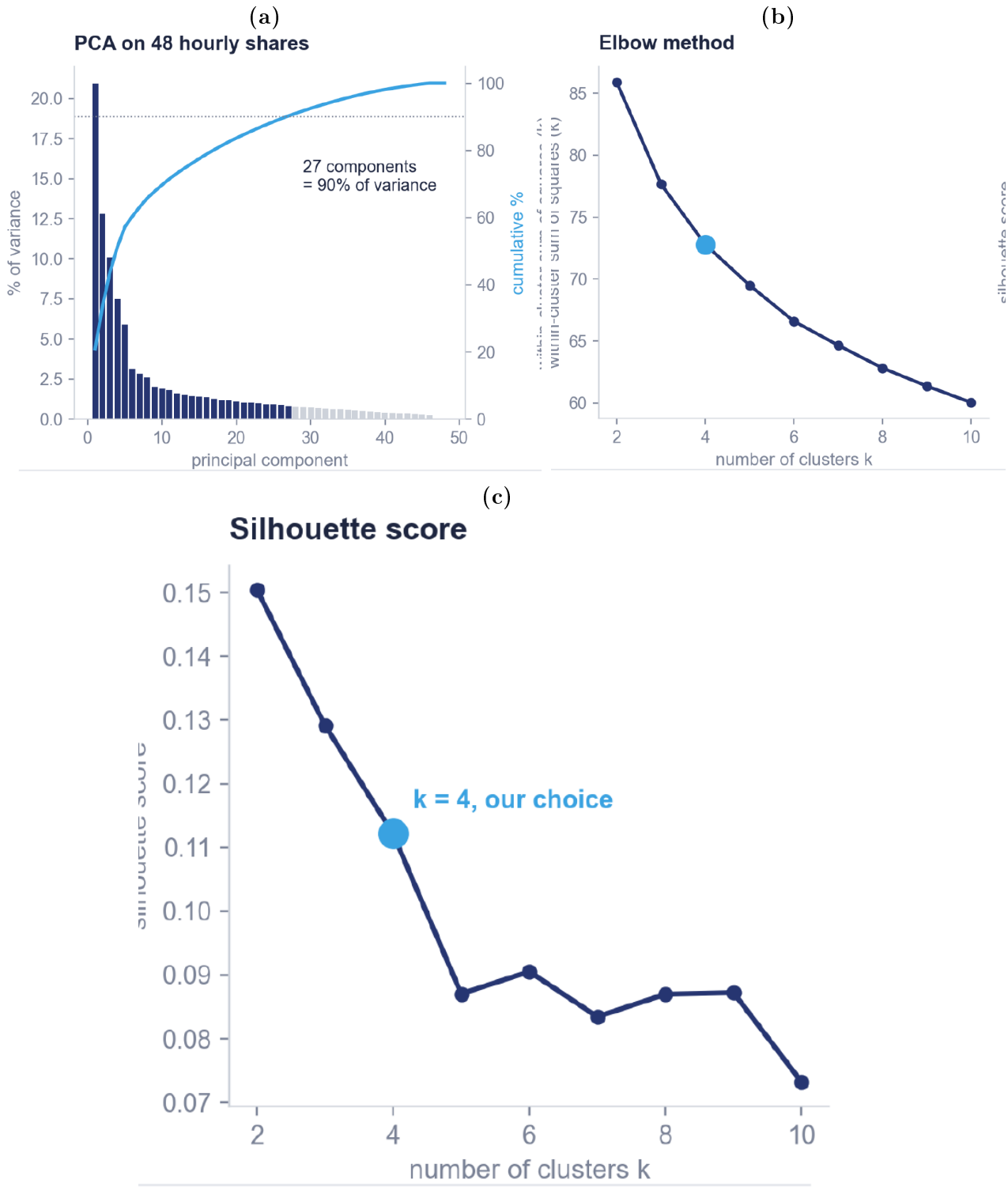


Figure 8: Choice of k. (a) PCA on 48 hourly shares (27 components reach 90% of the variance). (b) elbow method. (c) silhouette score for $k = 2$ to 10; $k = 4$ is our choice.

Stations therefore do not form natural, well-separated groups; they lie on a continuum. We chose $k = 4$ because four types can be named and each calls for a different action, whereas $k = 2$ would only separate "busy" from "quiet". The choice of four is a modelling decision.

3.2.3 Result

Figure 9 shows the average hourly profile of each type. Residential stations send out 8.7% of their weekday departures between 8 and 9am, and work stations receive 12.5% of their arrivals in the same hour. The mirror images (work stations in the evening, residential stations at night) complete the cycle.

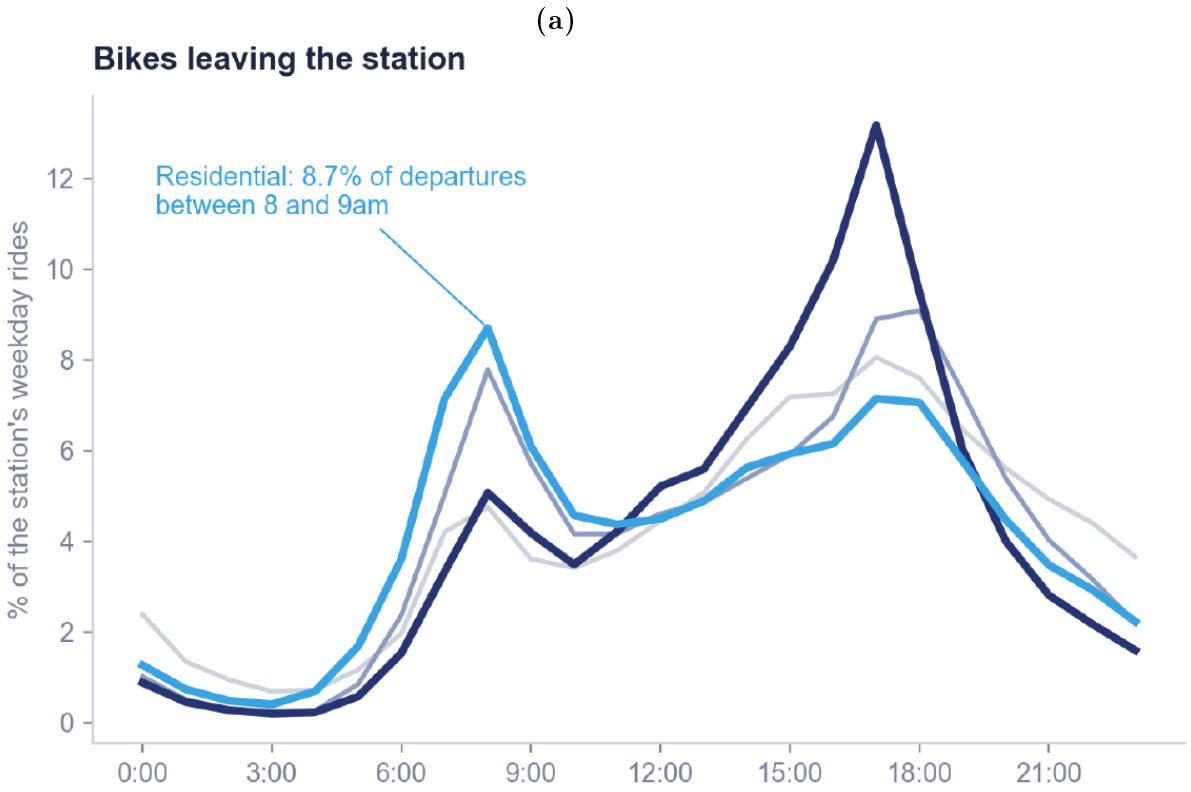


Figure 9: Average share of a station's weekday departures (a) and arrivals (b) in each hour, for the four types found by k-means.

(b)

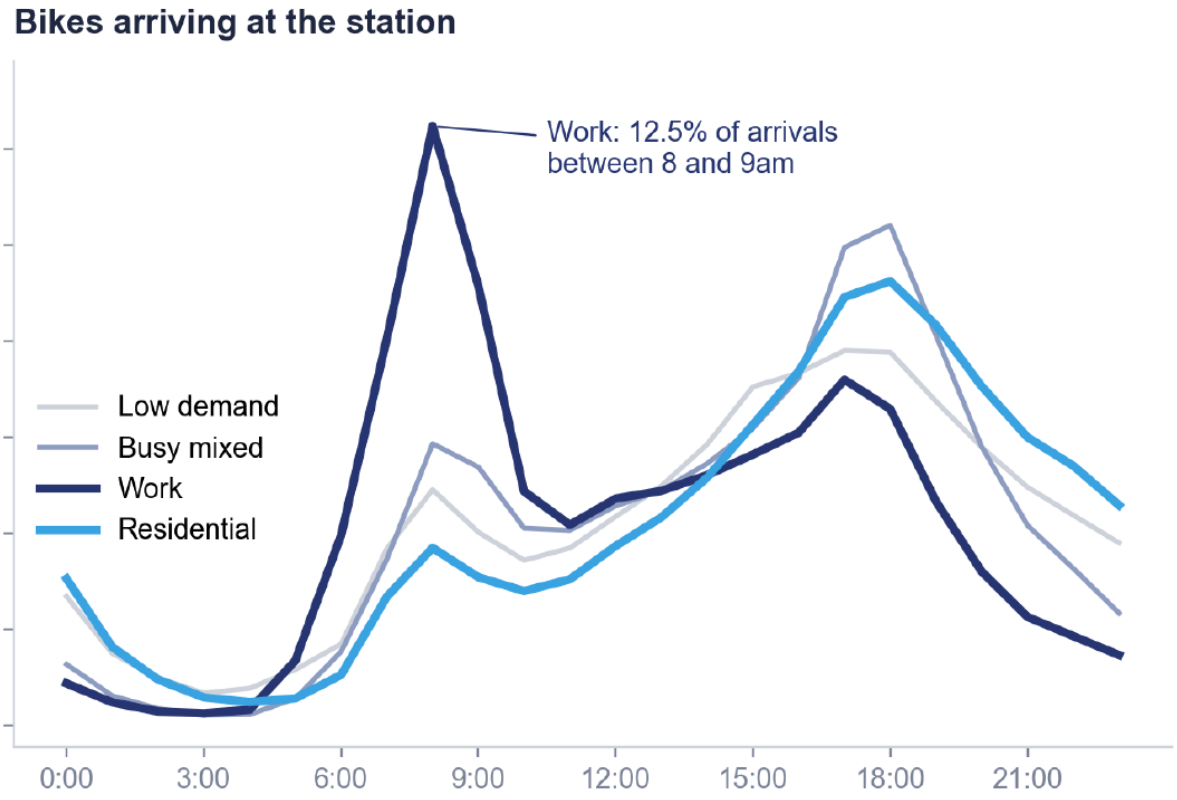


Figure 9: Panels continued.

Type	Main areas	Stations	Rides/day	Docks	Casual	Net flow/yr
Residential	upper Manhattan, Brooklyn, Queens	628	26	24	21%	−77
Work	Midtown, Financial District	337	79	43	20%	−10
Busy mixed	downtown, north Brooklyn	842	91	37	18%	+79
Low demand	outer areas	511	16	23	24%	−39

Table 5: The four station types (averages per station; rides per day over 396 days; net flow = rides ending − rides starting over the year).

What matters for the trucks is how often each type is empty or full during the day (Figure 10). On a weekday between 7 and 10am, residential stations are empty 6.4% of the time against 1.6% for work stations (busy mixed 3.3%, low demand 6%). Work stations are *full* 10.4% of the time in the morning, against 3.5% for residential stations. At night the cycle reverses: work stations are often empty and residential stations often full.

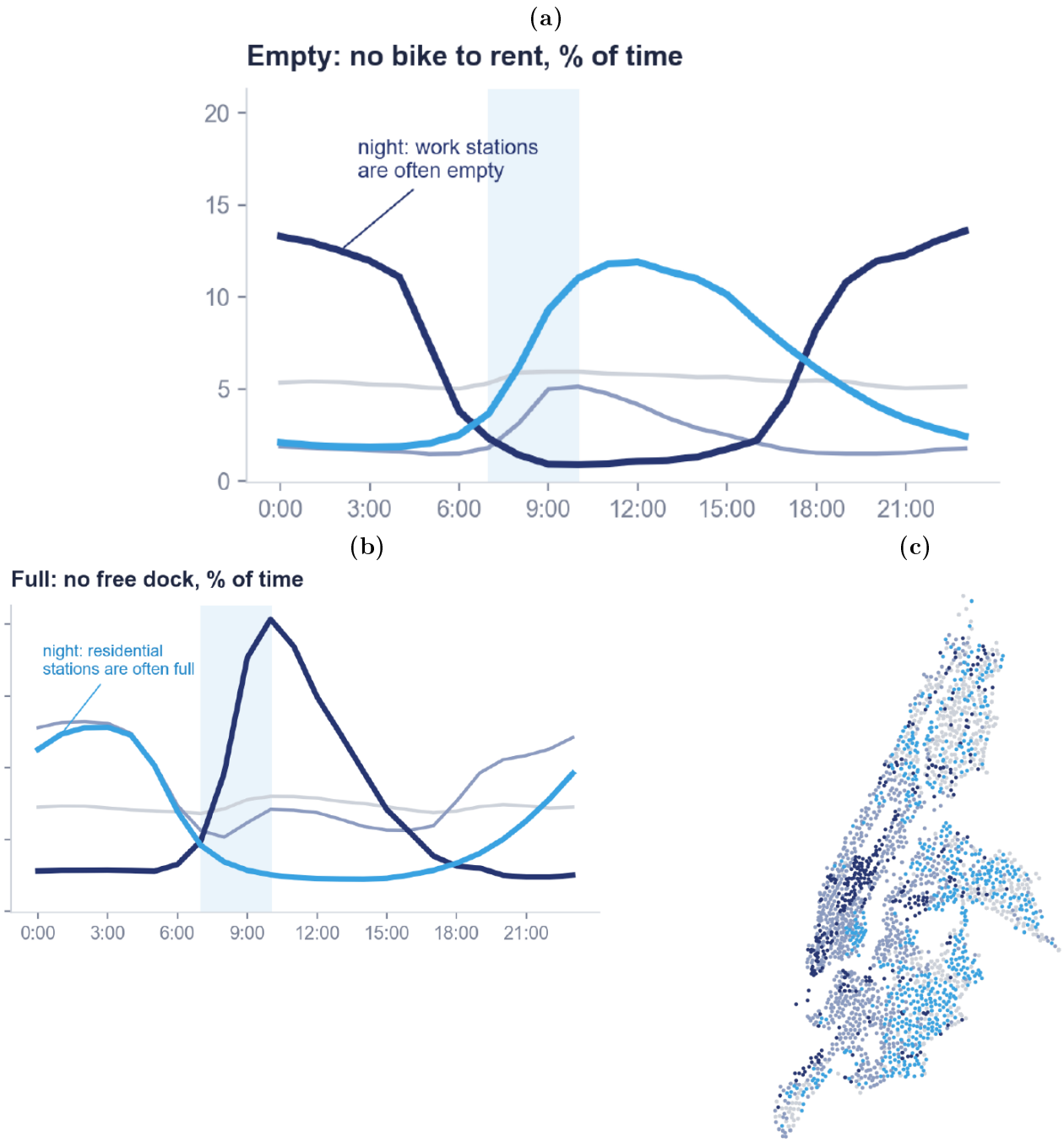


Figure 10: (a) Share of time stations of each type are empty (no bike to rent) during a weekday. (b) Share of time they are full (no free dock). (c) Map of the stations by type, with the colours of panels (a) and (b).

Key result

k-means only saw the hourly profiles, never the coordinates, yet the four types are geographically coherent (residential in upper Manhattan, Brooklyn and Queens; work in Midtown and the Financial District). The segmentation captures a real urban structure and not an artefact of the algorithm.

What this means for Citi Bike

The four types turn thousands of stations into four named operating policies. Residential stations lose bikes in the morning and need to be refilled *before* 7am. Work stations are the mirror image: they fill up in the morning, and at night they are rarely a good place to pick up bikes. Busy mixed stations gain bikes over the year (+79) and call for regular but limited intervention. Low-demand stations need few trucks but are also empty more often than their traffic suggests (6% of the weekday morning). The type is also a feature of the prediction model in Section 3.5.

The types describe *which* stations behave how. The next question is *how much* each station is used, and what moves that number.

3.3 Understanding the network II: what drives demand

If the operator knew tomorrow's demand at each station, it could anticipate shortages. We ask how much of a station's daily demand the calendar, the weather, the station itself and its recent history each explain.

3.3.1 Model

Daily departures per station are positive and right-skewed, so we model their logarithm:

$$\log(1 + D_{it}) = \beta_0 + \sum_j \beta_j x_{ijt} + \varepsilon_{it}. \tag{3}$$

The logarithm stabilises the variance (a busy station has larger absolute errors) and makes the coefficients *multiplicative*: a one-unit change in x_j changes expected departures by about $\exp(\beta_j) - 1$ in percent. We fit `LinearRegression` on 100,000 sampled station-days with an 80/20 split. The 26 explanatory variables belong to four groups: calendar (day of week, month, public holiday), weather (temperature, rain, wind), station (capacity, type) and the station's own history (rides a week earlier).

3.3.2 Fit

The test R^2 is **0.85** on the log scale (0.64 without the log), and the 5-fold cross-validated R^2 is 0.850, so the fit is not an accident of one split. Panel (a) of Figure 11 compares predicted and actual rides on 20,000 test station-days; panel (b) measures the contribution of each group of variables used alone.

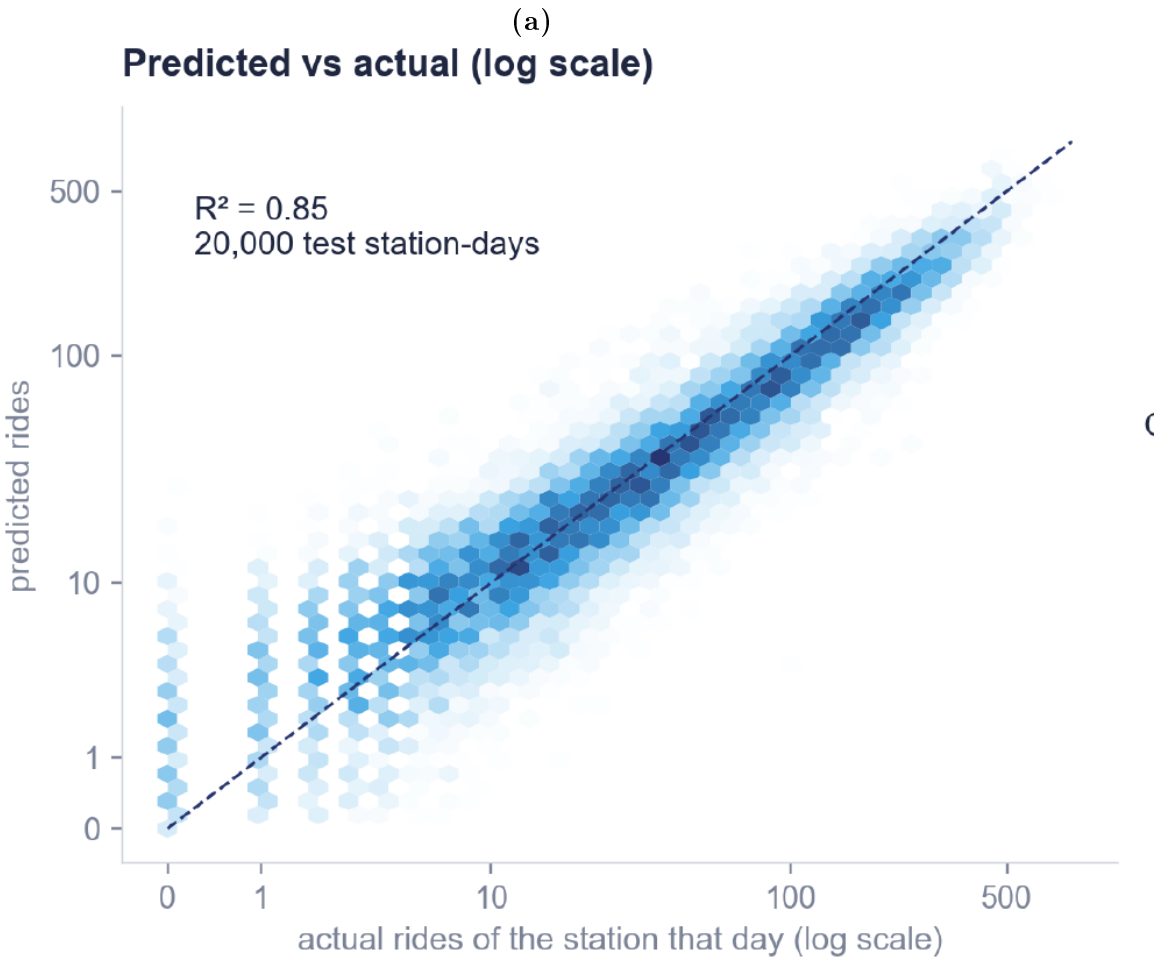


Figure 11: (a) predicted against actual rides on 20,000 test station-days (log scale). (b) out-of-sample R^2 with one group of variables at a time.

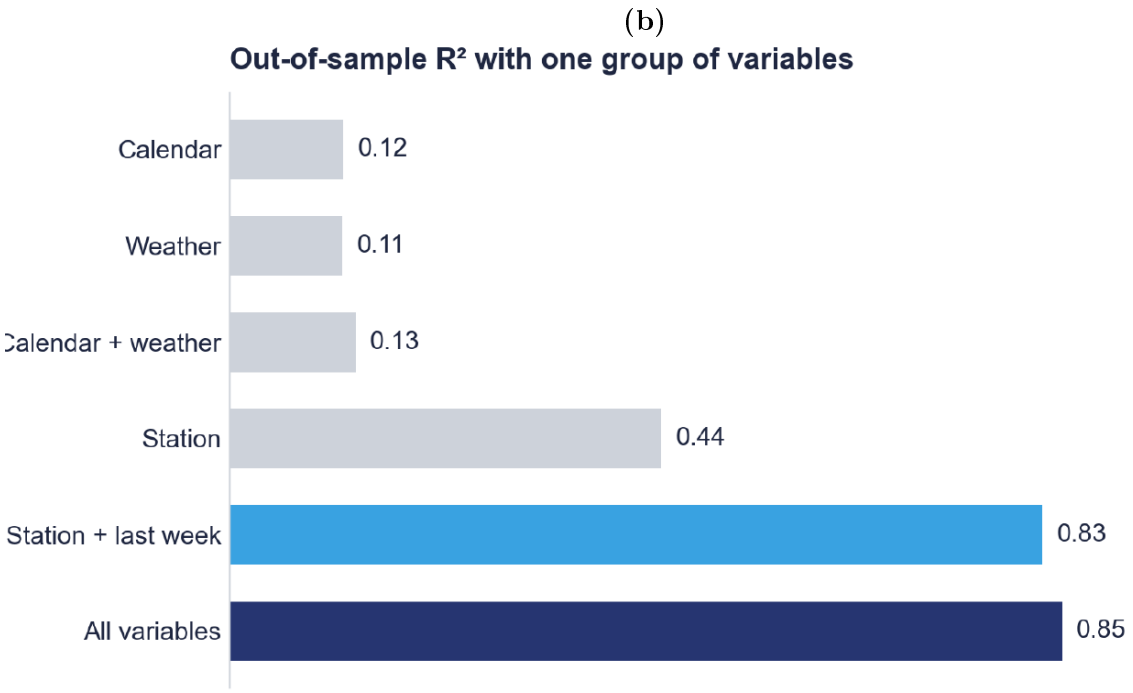


Figure 11: Panels continued.

The ordering is clear. The calendar alone explains 12% of the variation of a station's daily departures and weather alone 11%; the two together explain 13%. Knowing *which station* it is explains 44%, and adding last week's rides of the same station raises this to 83%. All variables together give 85%: past behaviour contains almost everything the other variables add.

3.3.3 Effects

Figure 12a translates coefficients into percentage effects using $\exp(\beta) - 1$. A public holiday cuts a station's rides by 23.8% (a coefficient of $\ln(1 - 0.238) \approx -0.272$), each degree adds 3.5% ($\beta \approx \ln 1.035 = 0.034$), each millimetre of rain removes 1.2% and each km/h of wind 0.6%. Saturdays and Sundays are 3.0% and 3.6% below Monday, and a station with 10% more rides a week earlier has 8.0% more rides today.

3.3.4 Regularisation and selection

With 26 correlated variables we ask which are needed. The Lasso minimises

$$\sum_i \left(y_i - \beta_0 - \textstyle\sum_j \beta_j x_{ij}\right)^2 + \alpha \textstyle\sum_j |\beta_j|, \tag{4}$$

which shrinks some coefficients to exactly zero. Figure 12b shows out-of-sample R^2 as the penalty falls: a single variable already gives about 0.78, and the curve is flat after 15 variables. At $\alpha = 0.01$ the Lasso keeps **15 of 26 variables with an R^2 of 0.847**, against 0.849 with all 26. A forward and backward stepwise selection (`mlxtend`, five variables) independently picks temperature, rain, capacity, last week's rides and March. Three methods point to the same short list of drivers.

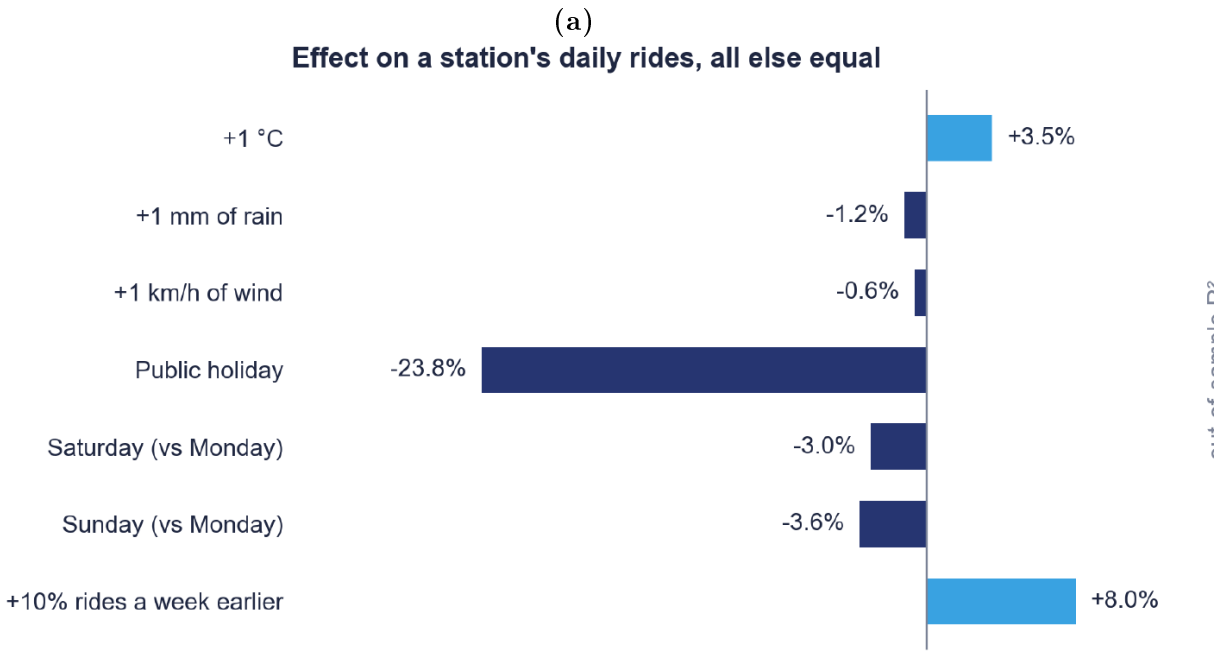


Figure 12: (a) effect of each variable on a station's daily rides in the full regression, as exp(coefficient) − 1. (b) Lasso with an increasing penalty (labels: α); $\alpha = 0.01$ keeps 15 of 26 variables.

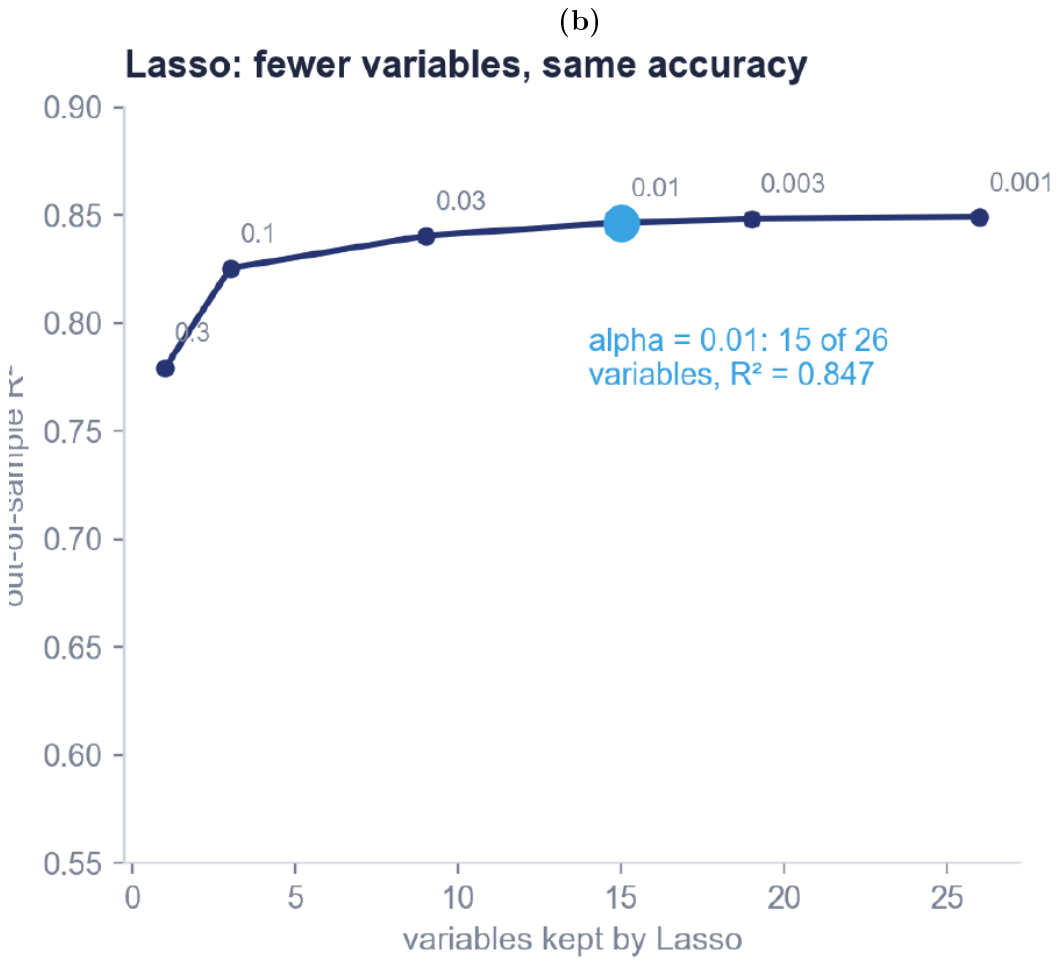


Figure 12: Panels continued.

3.3.5 City level versus station level

Hypothesis H1 expected time to matter more than the station. The result is more nuanced. For the *whole city*, with one row per day, weather alone gives $R^2 = 0.775$ and the day of the week alone $R^2 = 0.007$ (Figure 13). Weather sets the volume of the system: rides rise with temperature, fall with rain and, once rain exceeds 5 mm, the median day is slightly lower than a dry day. At the *station* level, however, it is the station's identity and recent history that matter. The weather moves all stations together; it says little about *which* station is short.

(a)

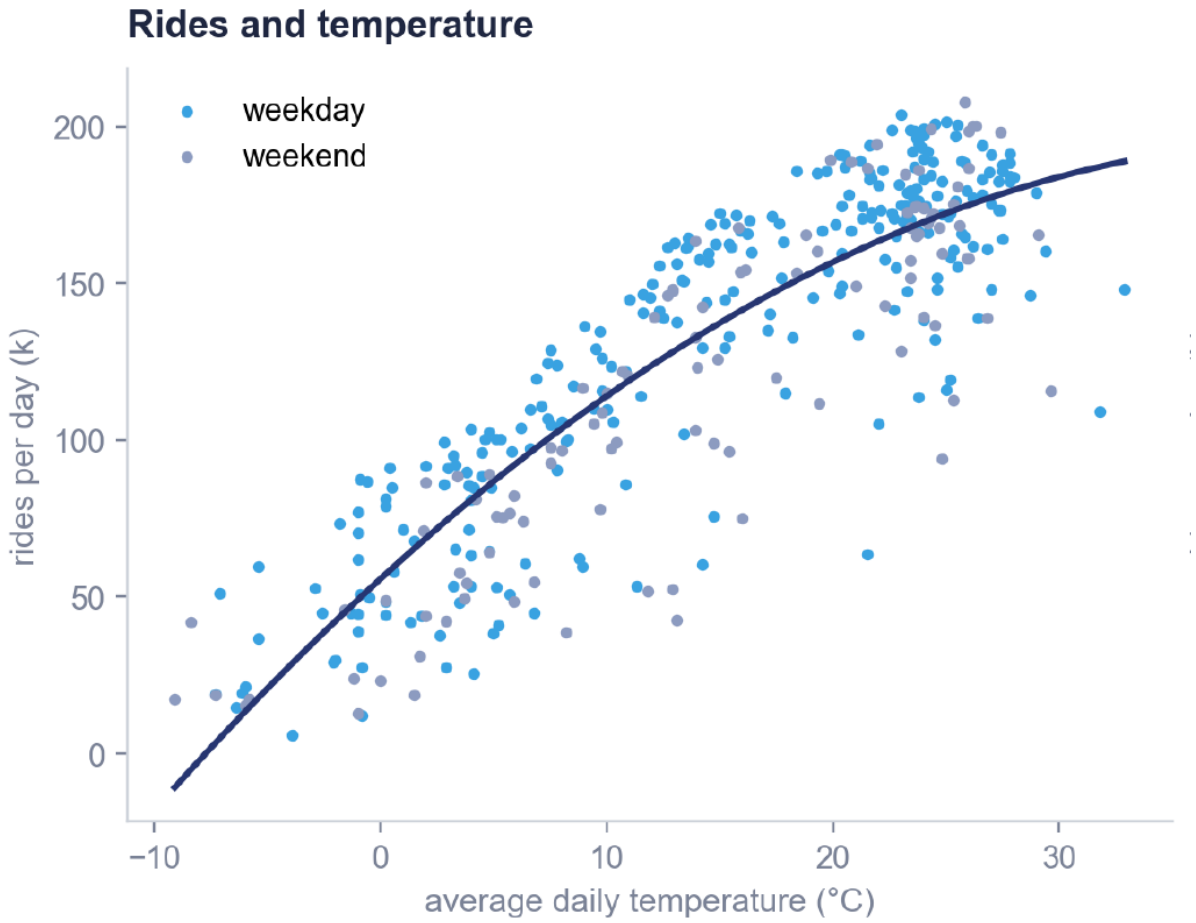


Figure 13: (a) rides per day against average daily temperature (one dot per day, quadratic fit for illustration). (b) rides per day by rain (230 dry days, 130 days with 0–5 mm, 22 days above 5 mm). (c) average rides per day (thousands) by day of the week.

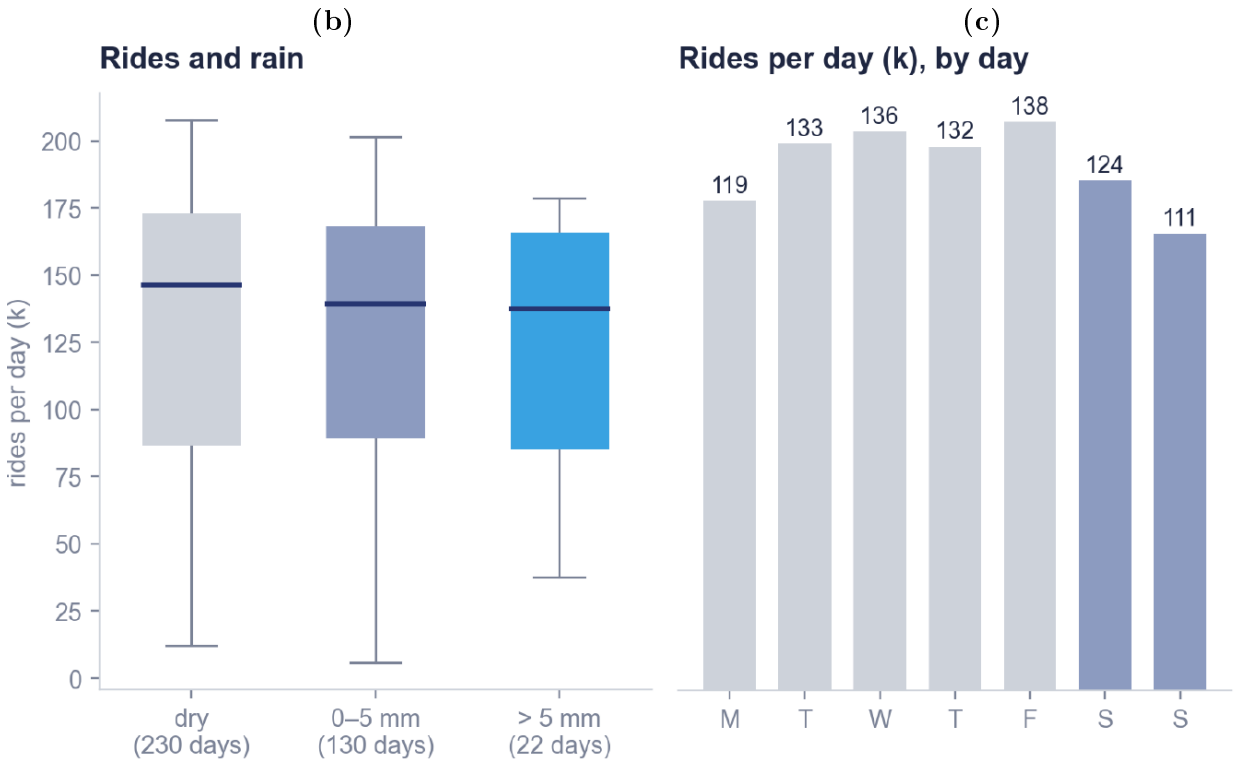


Figure 13: Panels continued.

What this means for Citi Bike

The city's volume can be forecast from the weather (78% of daily rides), which suits capacity planning: the number of trucks and drivers to schedule, with extra capacity on Mondays and in winter. By contrast, the *allocation* of those trucks cannot rely on the weather and must rely on each station's recent behaviour. That last week's rides explain 83% of the variation supports the approach of Section 3.5, in which the state of the station *today* is the main predictor of its state tomorrow.

Demand tells us how much stations are used. It does not tell us who uses them, which is the commercial side of the question.

3.4 Understanding the network III: who rides

Citi Bike's revenue depends on converting casual riders into members. We ask whether a ride can be identified as casual or member from its characteristics alone: duration, distance, hour, weekend, bike type, weather and station type. The test set has 40,036 rides, 18% of them casual.

3.4.1 Descriptive differences

Casual rides are longer (median 13 minutes against 8 for members) and more often at weekends (37% against 24%), on e-bikes (76% against 71%) and round trips (5% against 2%), but most casual rides are short (Figure 14). The clearest contrast concerns timing: **28% of casual rides** take place on weekdays at the commuting peaks (7–10am and 4–8pm) against **40% of member rides**.

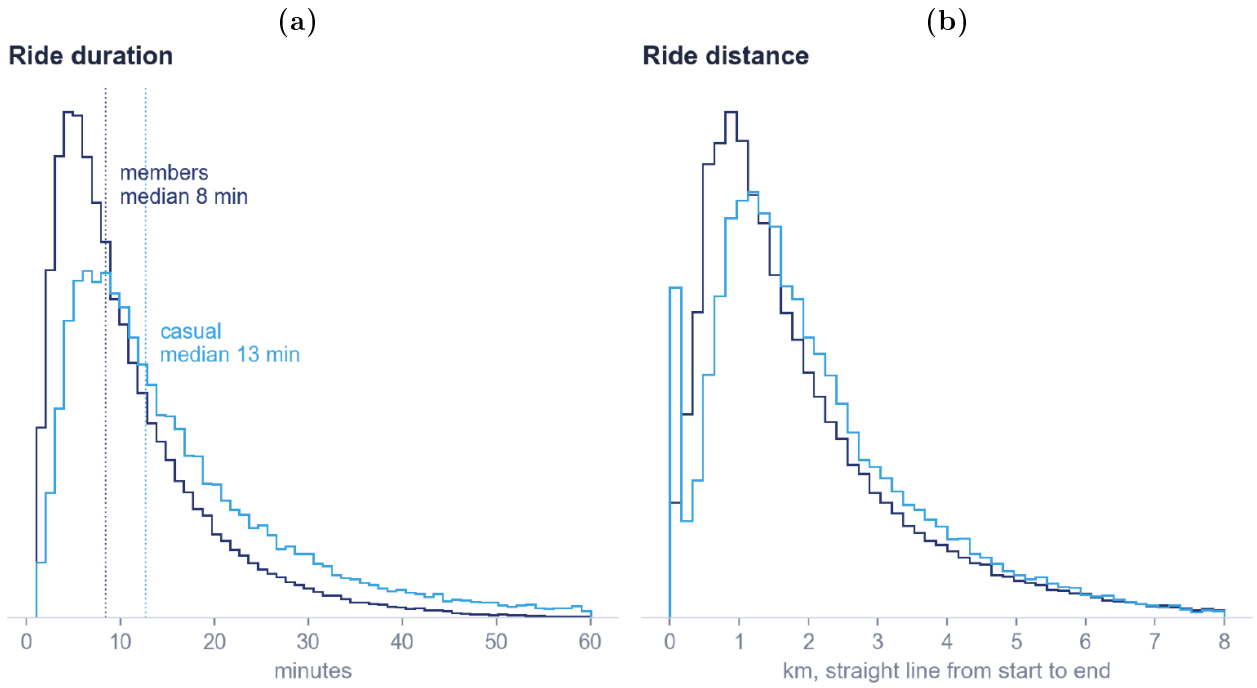


Figure 14: Members and casual riders compared on ride duration, ride distance (straight line from start to end) and three ride characteristics. Histograms normalised within each group; rides over 60 minutes and 8 km not shown. Dotted lines: medians.

(c)

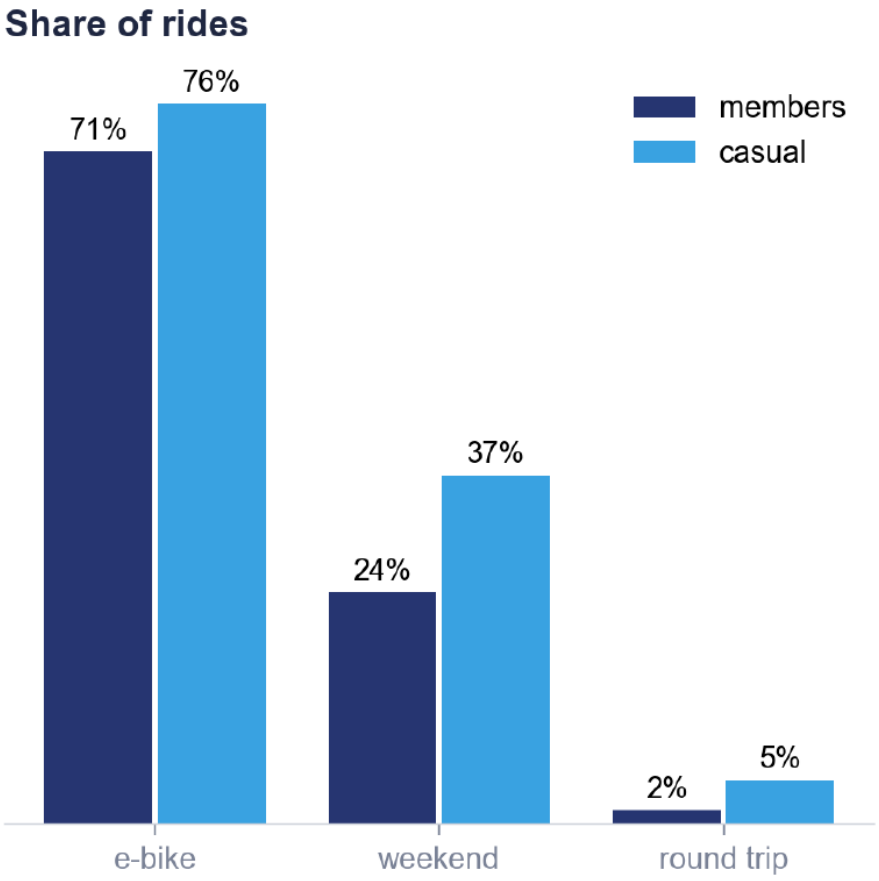


Figure 14: Panels continued.

3.4.2 Five classifiers

We compare five models on the same split: Naive Bayes (0.64), a decision tree (0.65), KNN (0.68 at its best k), logistic regression (0.69) and a random forest (0.72, 100 trees, `min_samples_leaf=20`). The ROC curves are in panel (a) of Figure 15.

KNN and the bias–variance trade-off. We use KNN to illustrate a central idea of model selection (Figure 15b). The classifier gives a ride the majority label among its k nearest neighbours in the space of standardised variables (trained on 20,000 rides; only k was varied, so this is an exploratory comparison). With a small k the model memorises the training rides: the training AUC is close to 1 but the test AUC is low, which is overfitting (low bias, high variance). With a large k it averages over too many neighbours and blurs real differences, which is underfitting (high bias, low variance). The test AUC peaks at $k = 101$ (0.68), between the two extremes. This k has nothing to do with the k of k-means: one is a number of neighbours, the other a number of clusters.

Random forest settings. The random forest averages many trees, each grown on a bootstrap sample. We control its complexity with the number of trees (100), the maximum depth, and `min_samples_leaf=20`, the minimum number of observations a leaf must contain. Requiring 20 observations per leaf prevents the trees from isolating individual rides and limits overfitting.

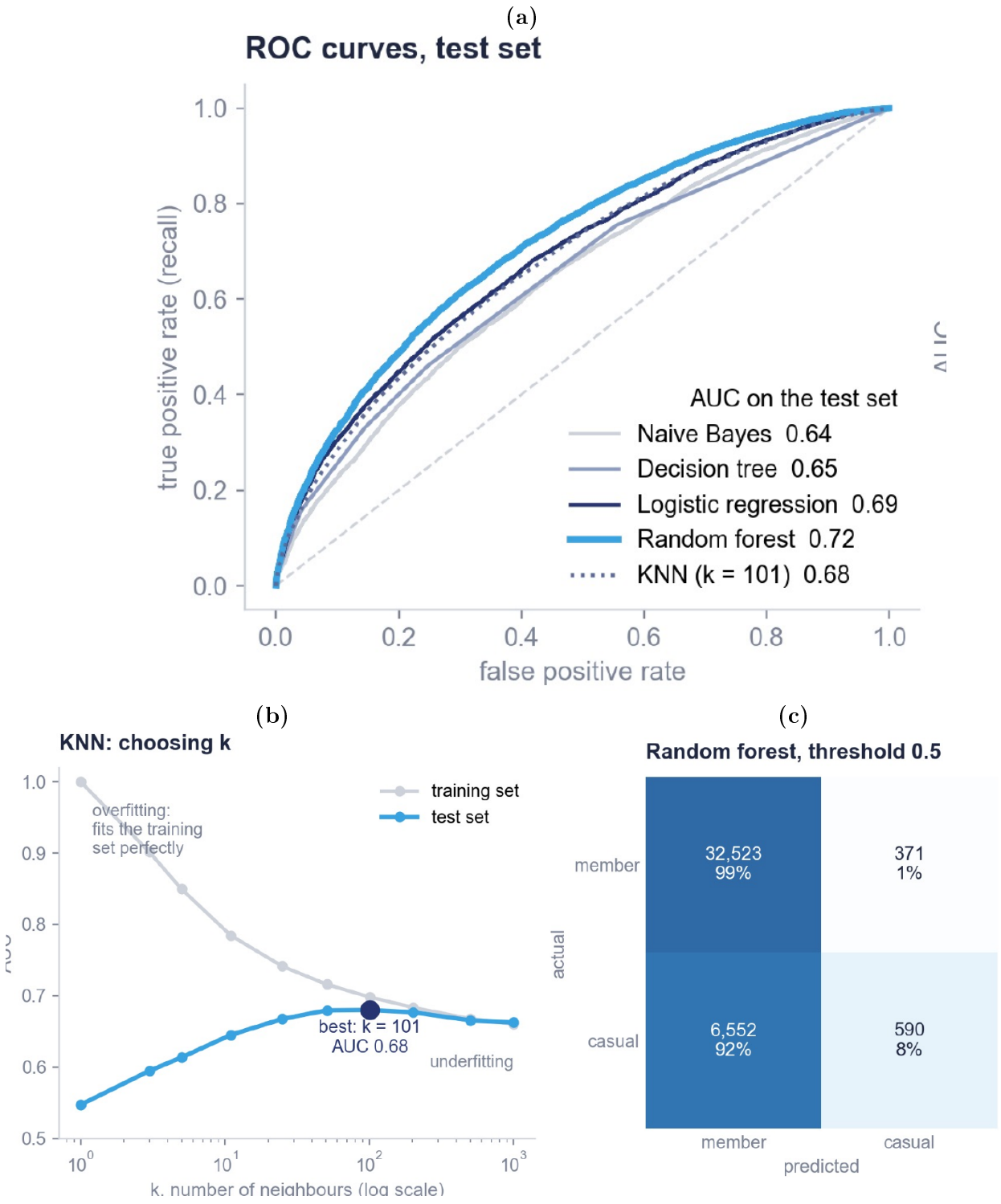


Figure 15: Member versus casual ride. (a) ROC curves on the test set. (b) KNN, AUC by number of neighbours k on the training and test sets. (c) confusion matrix of the random forest at threshold 0.5.

The most important variables of the forest are the duration and the distance of the ride, followed by the temperature and the hour of the day (Figure 16a). Whether the station is a work station matters least.

3.4.3 Accuracy is not the issue: imbalance is

At the default threshold of 0.5 the random forest labels 6,552 of the 7,142 casual test rides as member rides (confusion matrix in Figure 15): it finds only 590/7,142 ≈ 8% of casual riders. This does not make the model useless, since an AUC of 0.72 shows it ranks casual rides above member rides well beyond chance. It means that with an 18% minority class a threshold of 0.5 is too high. Resampling the training set raises recall. SMOTE synthesises new minority rides by interpolation, and random over-sampling duplicates them; both come from `imblearn` and are applied *only* to the training set, while the test set keeps the real 18% share.

For a logistic regression (Figure 16b), recall rises from 7% on the original training set to 38% with SMOTE and 55% with random over-sampling, while precision falls from 55% to 32% and 30% and accuracy from 82% to 75% and 69%. The F1 score, which combines the two, improves from 13% to 35% and 39%. The casual rides that the forest does recognise (590) tend to be concentrated around midday and the early afternoon rather than at the commuting peaks (Figure 16c).

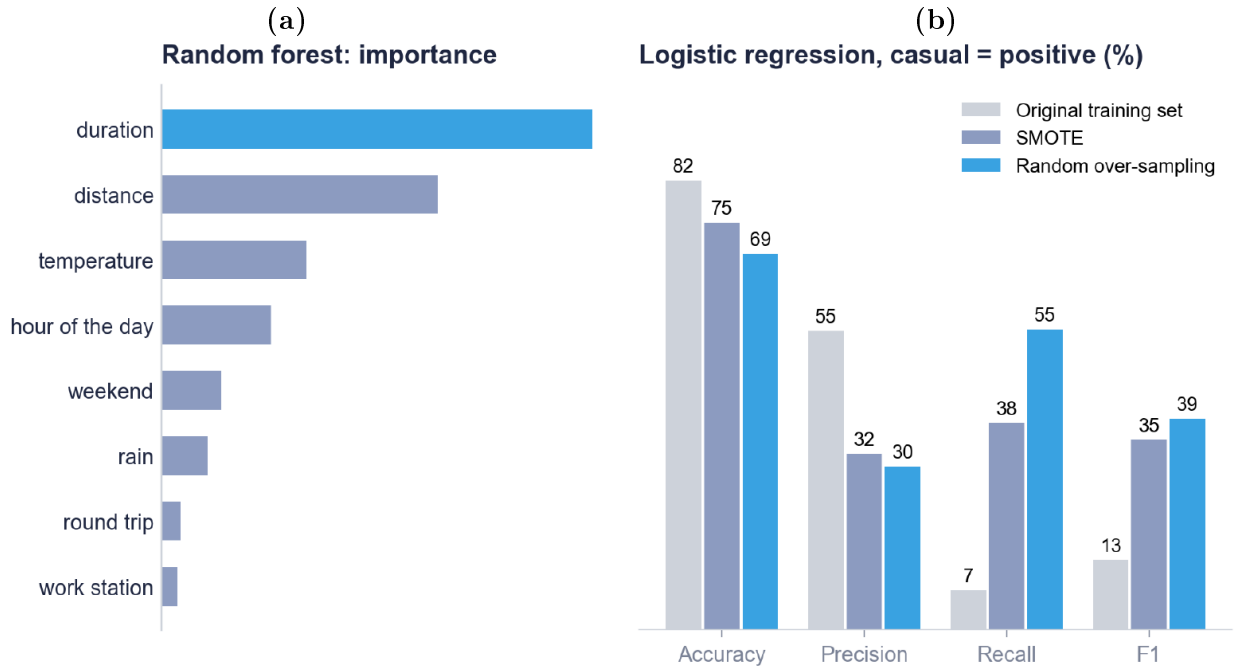


Figure 16: (a) random forest importance. (b) logistic regression (casual = positive) on the original, SMOTE and randomly over-sampled training sets. (c) share of weekday rides by hour for members, casual riders classified as members (6,552) and casual riders recognised (590).

(c)

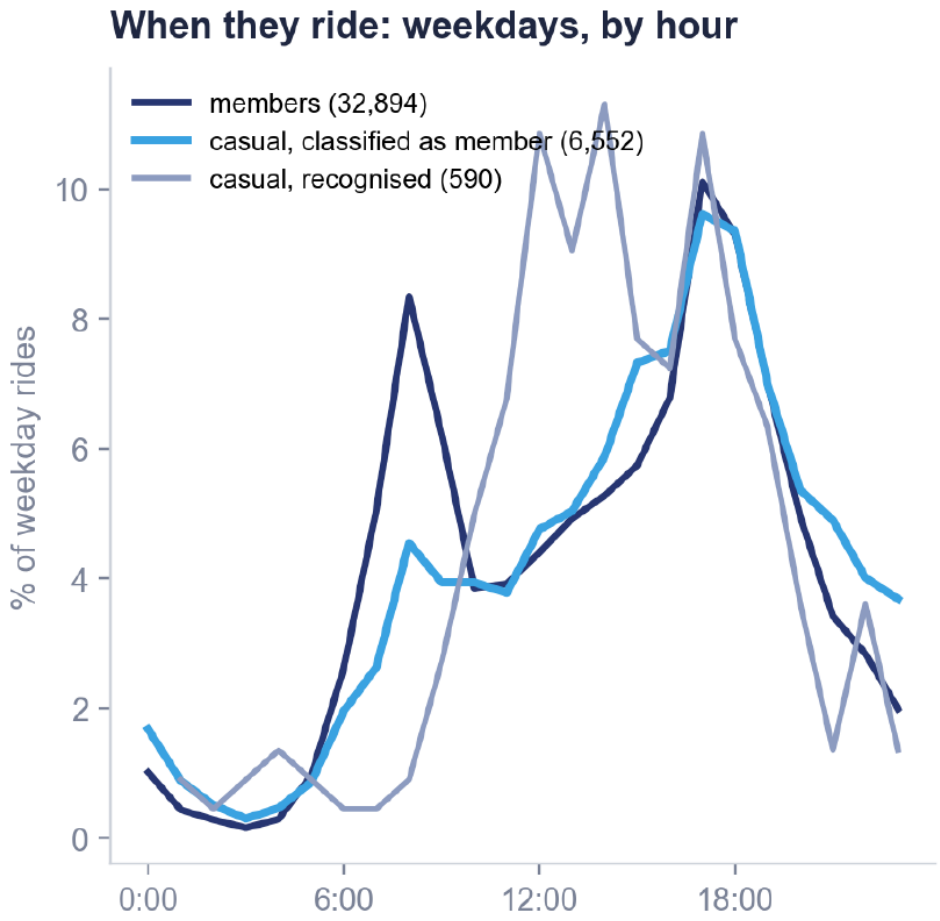


Figure 16: Panels continued.

Key result

Member and casual rides overlap strongly: the best AUC is 0.72. Hypothesis H2 is therefore only partly supported. Casual riders are not a separate group; many of them ride like commuters.

What this means for Citi Bike

This overlap is the commercial opportunity. A casual rider who rides at 8am on weekdays behaves like a member and may be a good candidate for a membership. These riders are the natural target of an offer: 28% of casual rides are in this situation. The classifier should not label individual rides; the offer should target *customers* who repeatedly ride at commuting peaks, and an A/B test would measure the conversion. Because the classifier is weak at the ride level, aggregating many rides per customer is also statistically sounder.

We now have a map of the network and of its riders. The central question remains: which stations will be empty tomorrow morning?

3.5 Predicting tomorrow morning

We define a prediction problem that respects what is known at night, compare four models, test the best one against a rule that needs no model, and turn its scores into two lists for the trucks.

3.5.1 Problem design

Figure 17 summarises the design.

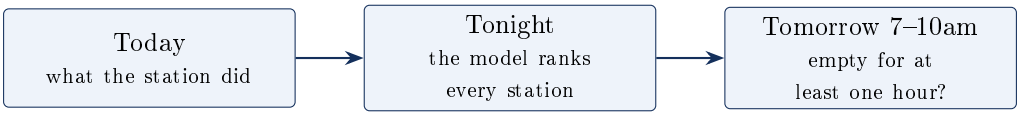


Figure 17: Timing of the prediction. Inputs use only information available the evening before.

Target. $y_{it} = 1$ if station i is empty in at least 33% of the 7–10am snapshots of day $t + 1$ (at least one hour). This happens on 4.7% of station-days; panel (b) of Figure 18 shows how the threshold cuts the distribution of the share of the morning that stations spend empty (log scale).

Inputs, all known tonight.

- the station's state today: share of the morning and of the evening it was empty, share it was full;
- today's rides: departures, arrivals, morning net flow;
- the station: capacity and type from the clustering;
- tomorrow's calendar: day of the week, weekend, public holiday;
- tomorrow's weather: average temperature and rain.

Observed weather replaces the forecast; since removing the weather variables leaves the AUC essentially unchanged (Section 3.5.6), this simplification does not drive the results.

Design of the test. We sample 150,000 station-days and split them 80/20: 120,000 for training (with 5-fold cross-validation) and a test set of 30,000, never used for training, in which 4.6% of the station-days are empty the next morning. Four models are compared, with the AUC as main metric.

3.5.2 Why yesterday predicts tomorrow

A station that was empty this morning is often empty again tomorrow (Figure 18a). A station never empty this morning (89% of station-days) has a **2.7%** chance of a shortage tomorrow. The chance rises to 8.3% if it was empty up to 10% of this morning, 12.8% for 10–33%, 25.6% for 33–66% and **42.3%** if it was empty more than two thirds of this morning, a ratio of about 16 between the two extremes. This persistence is the main predictive signal, and it fits the structure found in Section 3.3: stations have stable commuting roles.

(a)

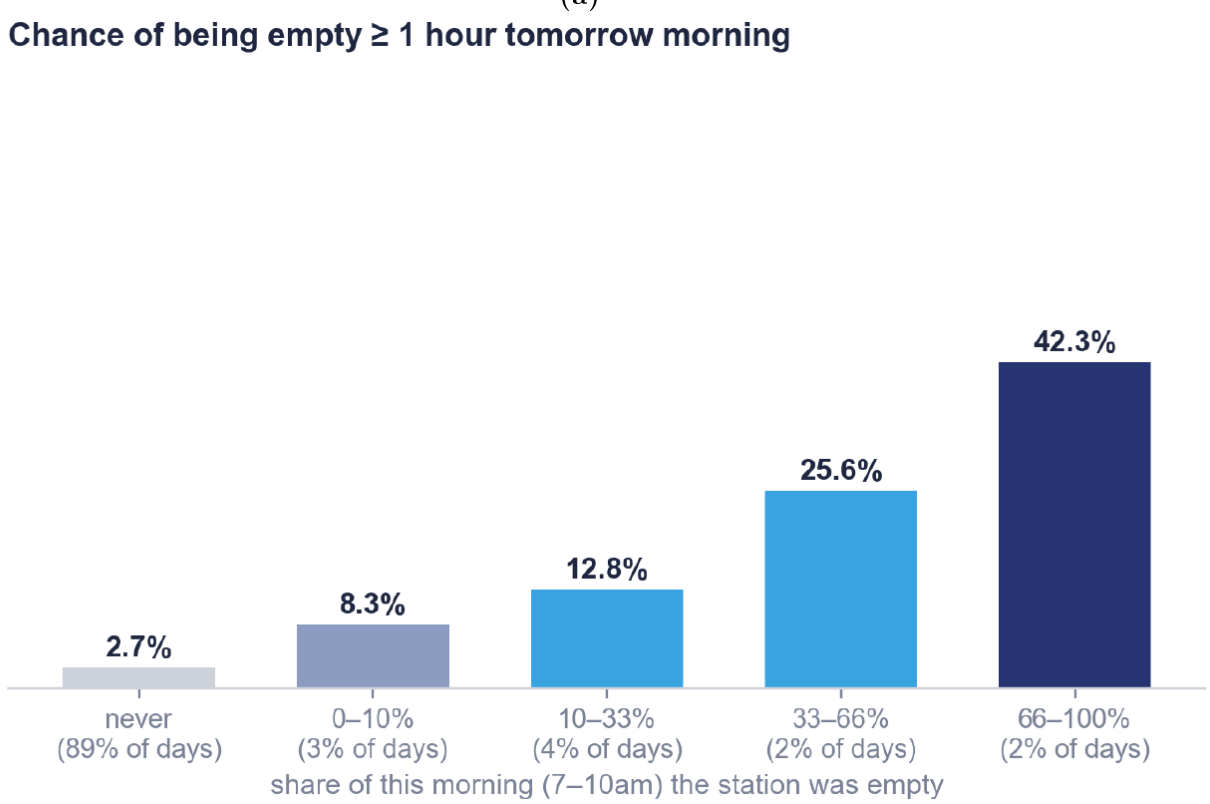


Figure 18: (a) chance of being empty for at least an hour tomorrow, by the share of this morning (7–10am) the station was empty (in brackets: share of all station-days). (b) how the target is defined (log scale; blue = station-days counted as empty).

(b)

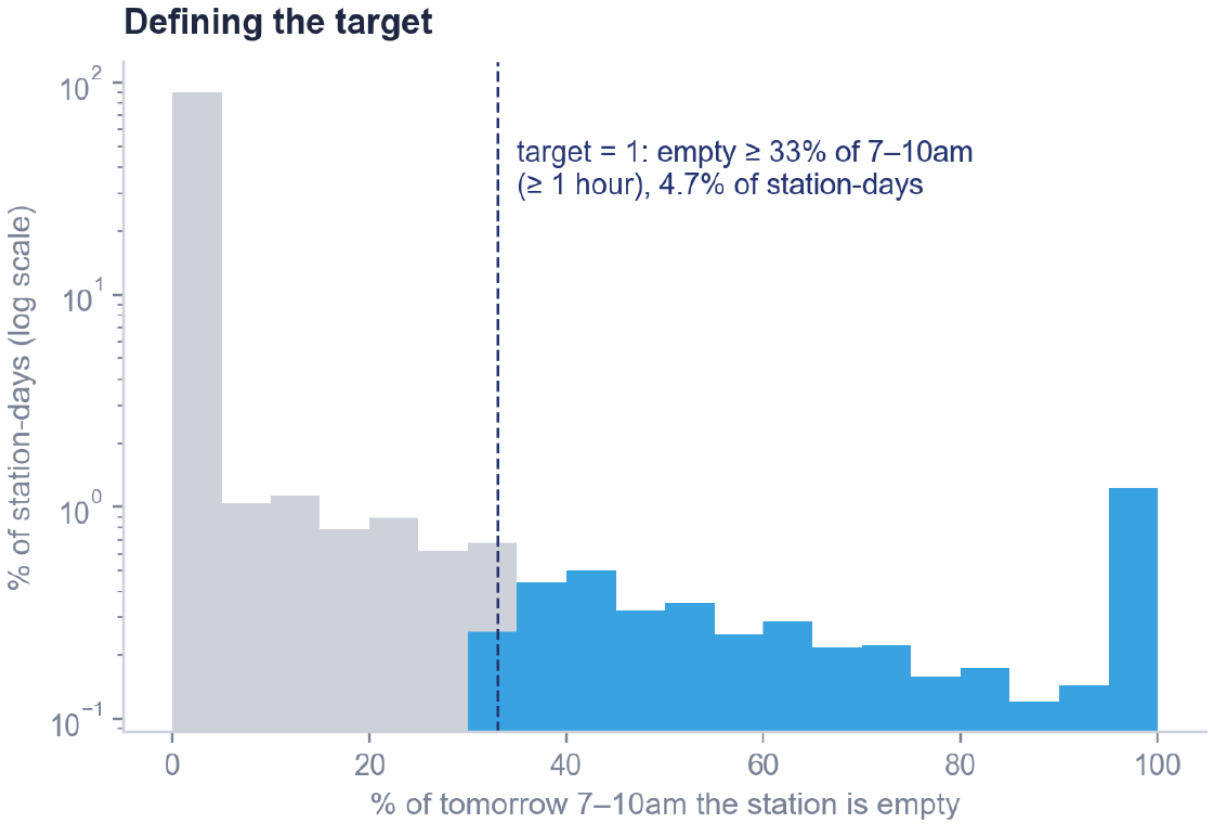


Figure 18: Panels continued.

3.5.3 Models and results

We compare a decision tree (`max_depth=5`), a logistic regression, a random forest (100 trees, `min_samples_leaf=20`) and XGBoost (`max_depth=4`, learning rate $\eta = 0.3$, 100 boosting rounds). XGBoost builds trees one after another, each correcting the errors of the previous ones by minimising the regularised objective

$$\mathcal{L} = \sum_i \ell\big(y_i, \hat{y}_i\big) + \sum_k \Omega(f_k), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2, \tag{5}$$

where ℓ is the logistic loss and Ω penalises the number of leaves T and the size of the leaf weights w.

Model	Test AUC	5-fold CV AUC
Decision tree (depth 5)	0.842	0.842
Logistic regression	0.865	0.860
Random forest	0.893	0.891
XGBoost	**0.906**	**0.899**

Table 6: Shortage models: AUC on the 30,000 test station-days and mean AUC in 5-fold cross-validation on the training set. Five other random splits give an XGBoost AUC of 0.896–0.903.

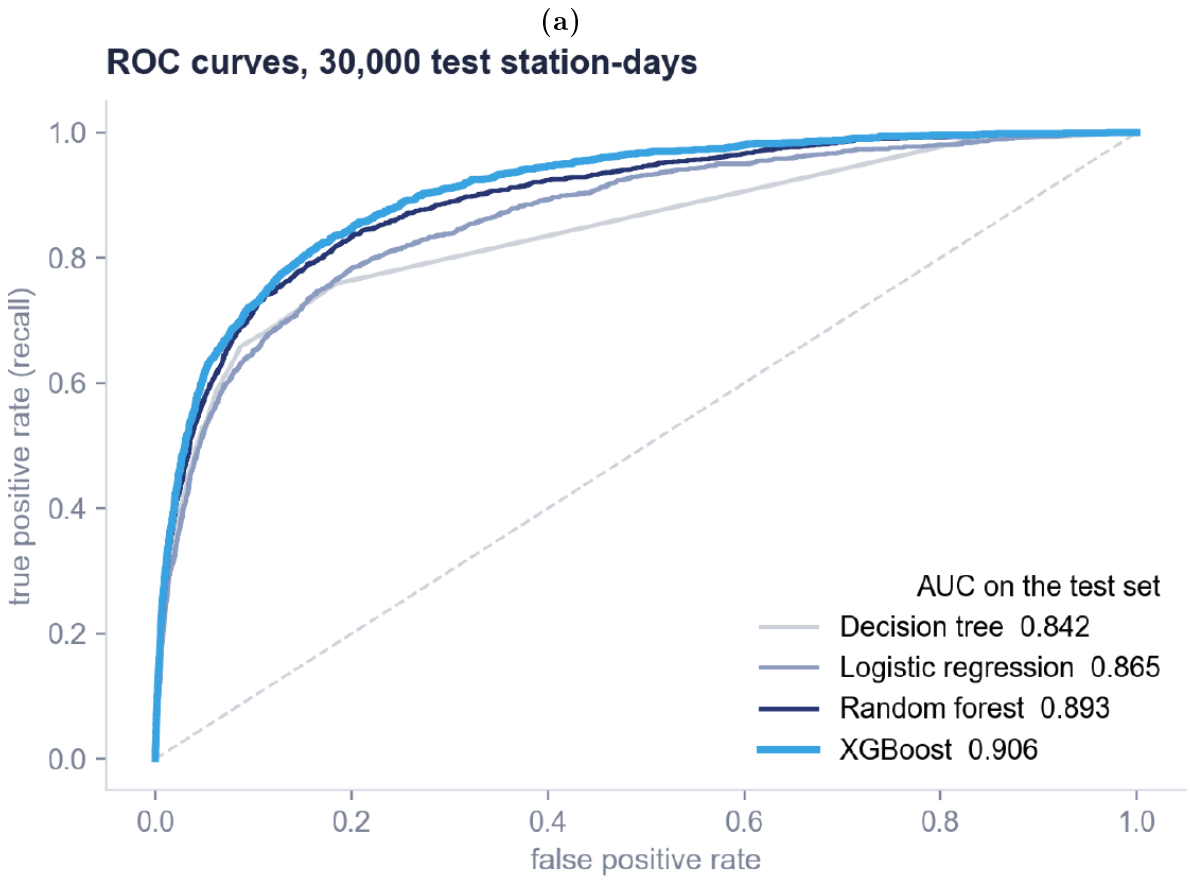


Figure 19: (a) ROC curves of the four models on the 30,000 test station-days. (b) AUC in 5-fold cross-validation on the training set (dot: mean ± 2 standard deviations; cross: test set).

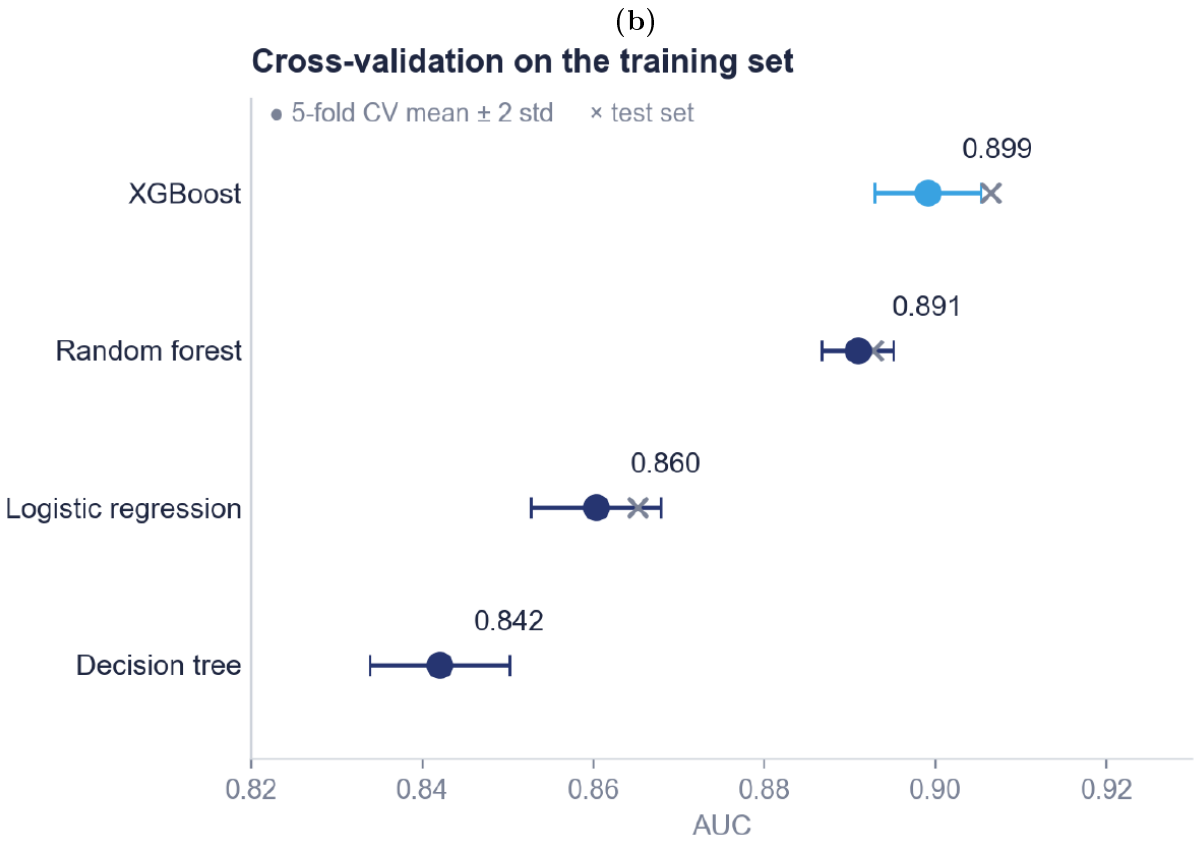


Figure 19: Panels continued.

The tree-based ensembles beat the linear model, which suggests nonlinear effects: a station empty for most of the morning is much more at risk than the average of its history suggests. XGBoost is the best model, and the small gap between its test AUC (0.906) and its cross-validated AUC (0.899) indicates little overfitting.

3.5.4 A test against the operator's current practice

An AUC measures ranking quality over all pairs of stations. A nightly list is much shorter than that, so we also test the model against a practical *benchmark rule*: visit the stations that were empty longest today, with the same number of stops. To avoid any leakage, the split here is **by date**: 80% of the nights to train and the last 20% (**77 nights** the model never saw) to test.

With 10% of the stations, the model catches **62%** of shortages (hit rate 31%), the rule 51% (25%) and a random list 10% (4.9%). The model also beats a second rule, "empty longest this evening" (48%). The AUC on the test nights is 0.905 for the model and 0.855 for the rule. The gain holds at other list sizes (Figure 20a: 43% against 33% at 5%, and 79% against 76% at 20%), and it is not a feature of a few lucky nights: the model is better than the rule on 68 of the 77 nights (Figure 20b).

(a)

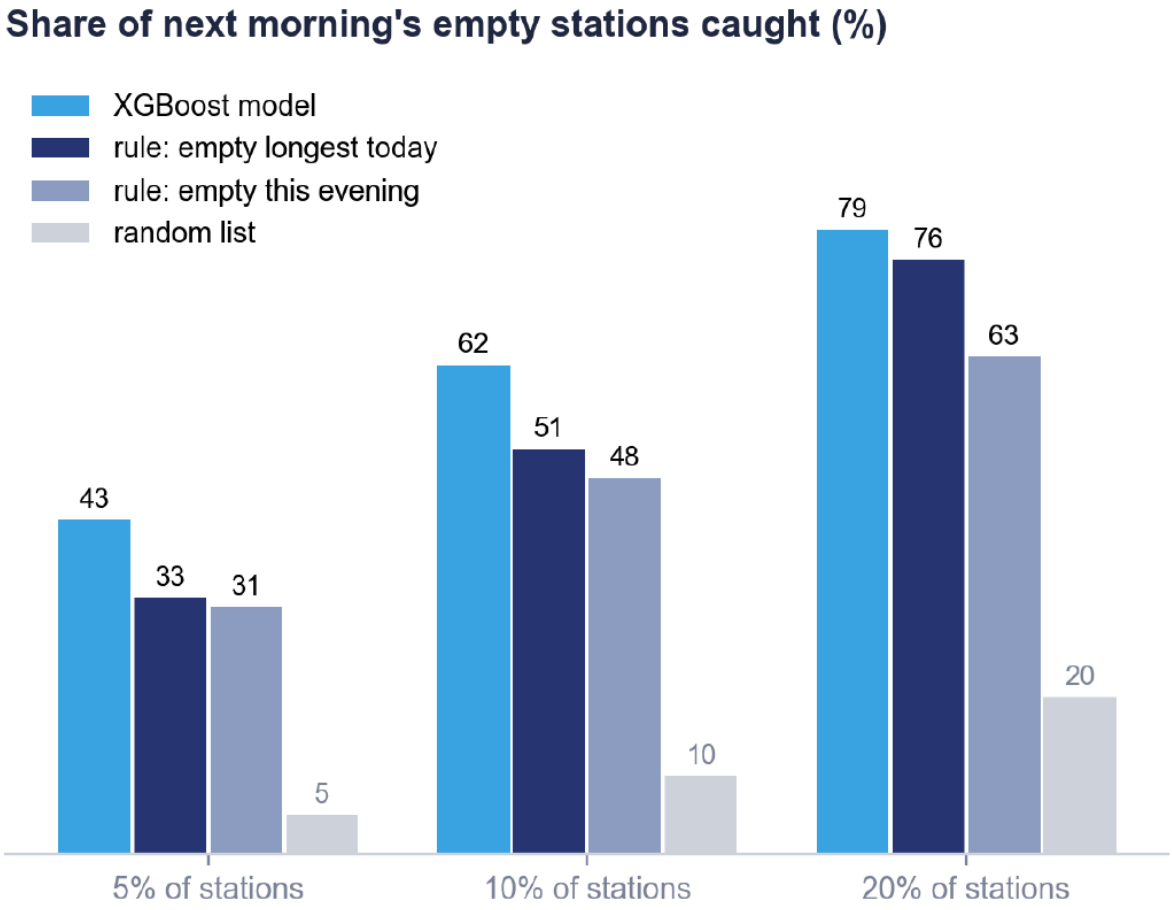


Figure 20: (a) share of next morning's empty stations caught, by list size, for the XGBoost model, two simple rules and a random list. (b) share caught by the 10% list on each of the 77 test nights, model against the rule "empty longest today".

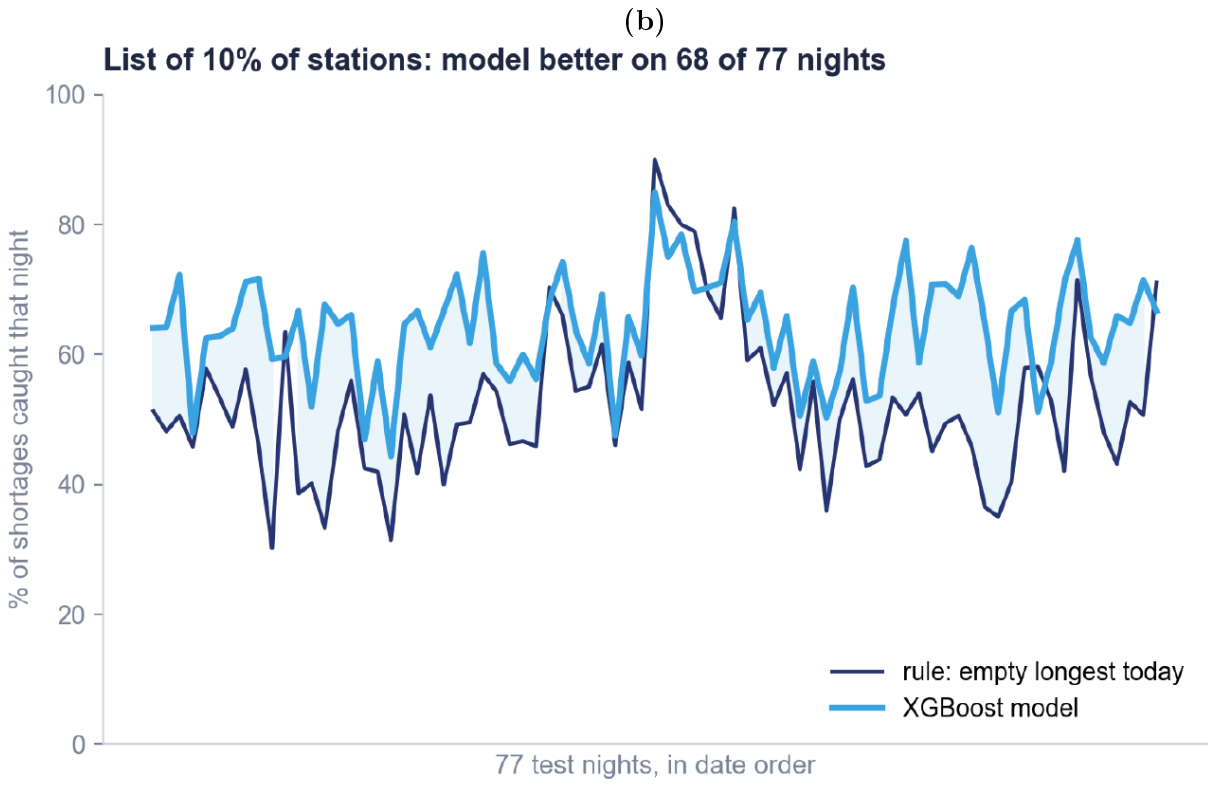


Figure 20: Panels continued.

What it means in stops. Our table has about 2,150 station-days per day (851,613/396), so a 10% list has about 215 stops, and the base rate of 4.7% implies about 100 shortages a night. With the model, about 62 of them are on the list; with the rule, about 51. The model therefore finds roughly 11 additional shortages a night for the same number of stops. (These orders of magnitude are our own computation from the figures above, not an additional output of the model.)

On the random test set (Figure 21), the 10% list catches 67% of empty stations, the 5% list 49% and the 20% list 82%. The hit rate falls as the list grows (45%, 31%, 19% and 14% for the top 5, 10, 20 and 30%), against a random list at 4.6%. The two views are consistent: $0.31 \times 10\%/4.6\% \approx 67\%$.

(a)

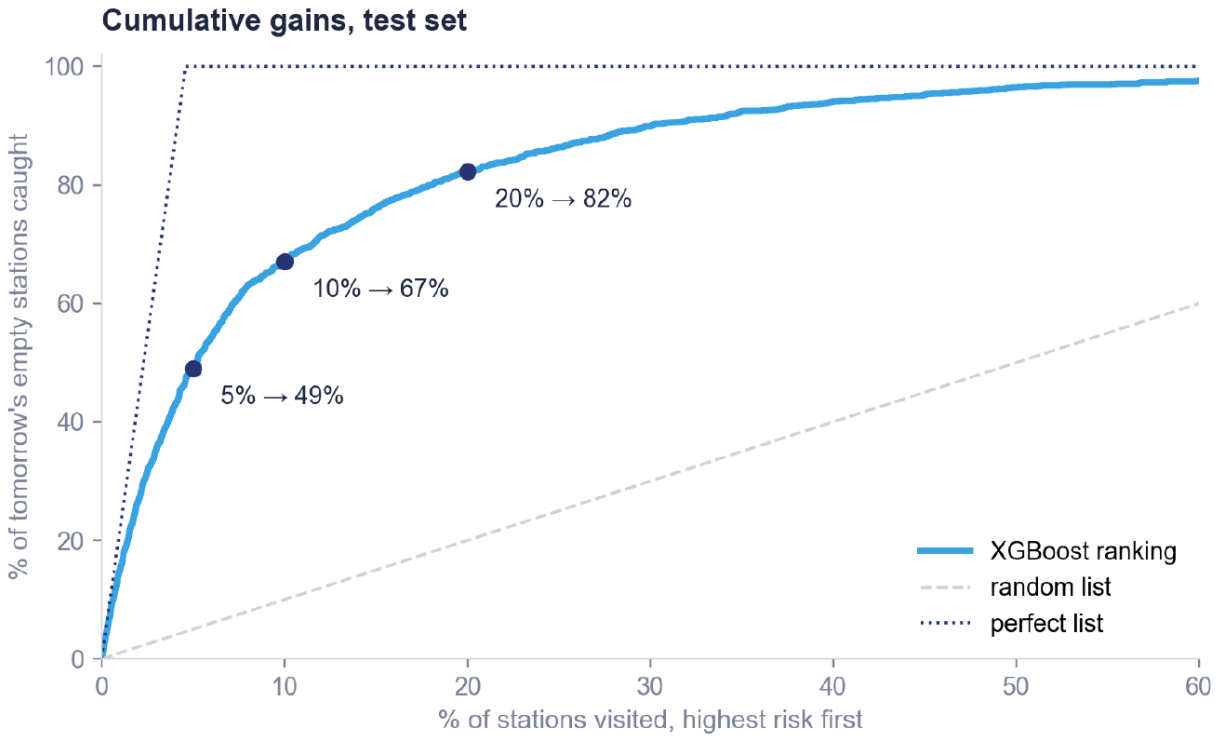


Figure 21: (a) cumulative gains, share of tomorrow's empty stations caught when visiting stations in order of predicted risk (random test set of 30,000 station-days). (b) hit rate by list size; the dashed line is a random list (4.6%).

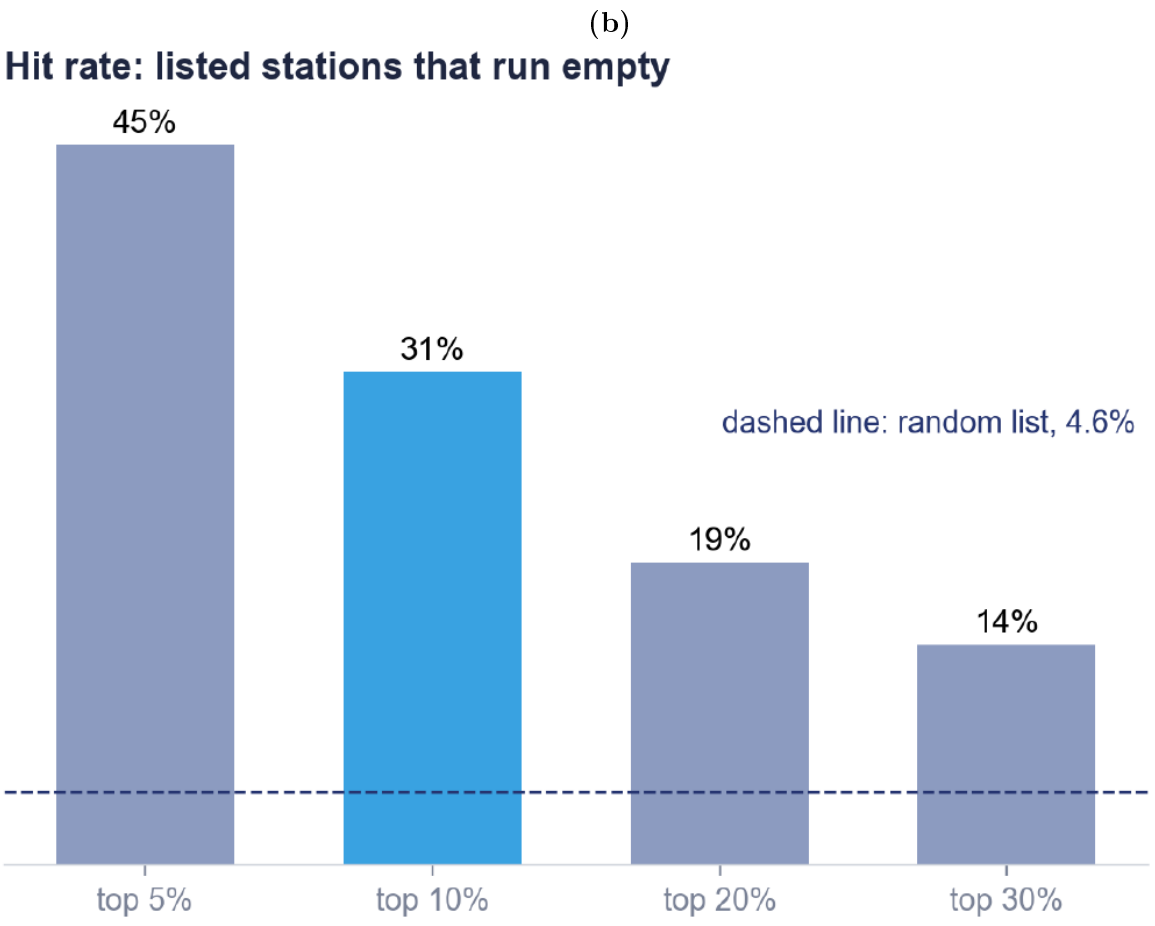


Figure 21: Panels continued.

List size (random test set)	5%	10%	20%	30%
Hit rate	45%	31%	19%	14%
Share of shortages caught	49%	67%	82%	–
Lift over a random list (hit rate / 4.6%)	9.8	6.7	4.1	3.0

Table 7: Cumulative gains on the random test set.

3.5.5 Threshold and confusion matrix

At the default threshold of 0.5, XGBoost has a precision of 63% and a recall of only 24%: it labels a station as empty only when it is quite sure (Figure 22). On the 30,000 test station-days it finds 336 of the 1,377 empty stations, misses 1,041 and raises 198 false alarms. For the trucks this is the wrong operating point. The dispatcher does not need a yes or no for each station but a ranking from which to take the first stops. We therefore rank instead of thresholding: the top 10% of stations (probability of at least about 0.10) have a precision of about 31%, about 6.6 times the 4.7% base rate, and a recall of about 67%.

(a)

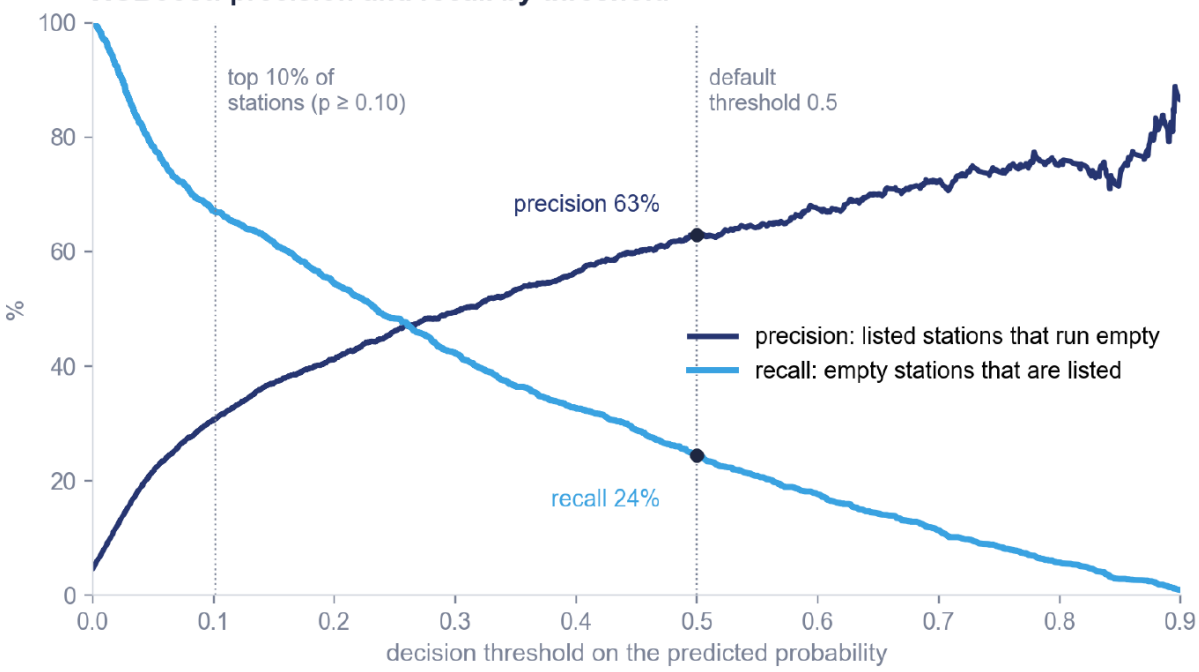


Figure 22: (a) precision and recall of XGBoost for every decision threshold. (b) confusion matrix at the default threshold of 0.5 (test set of 30,000 station-days).

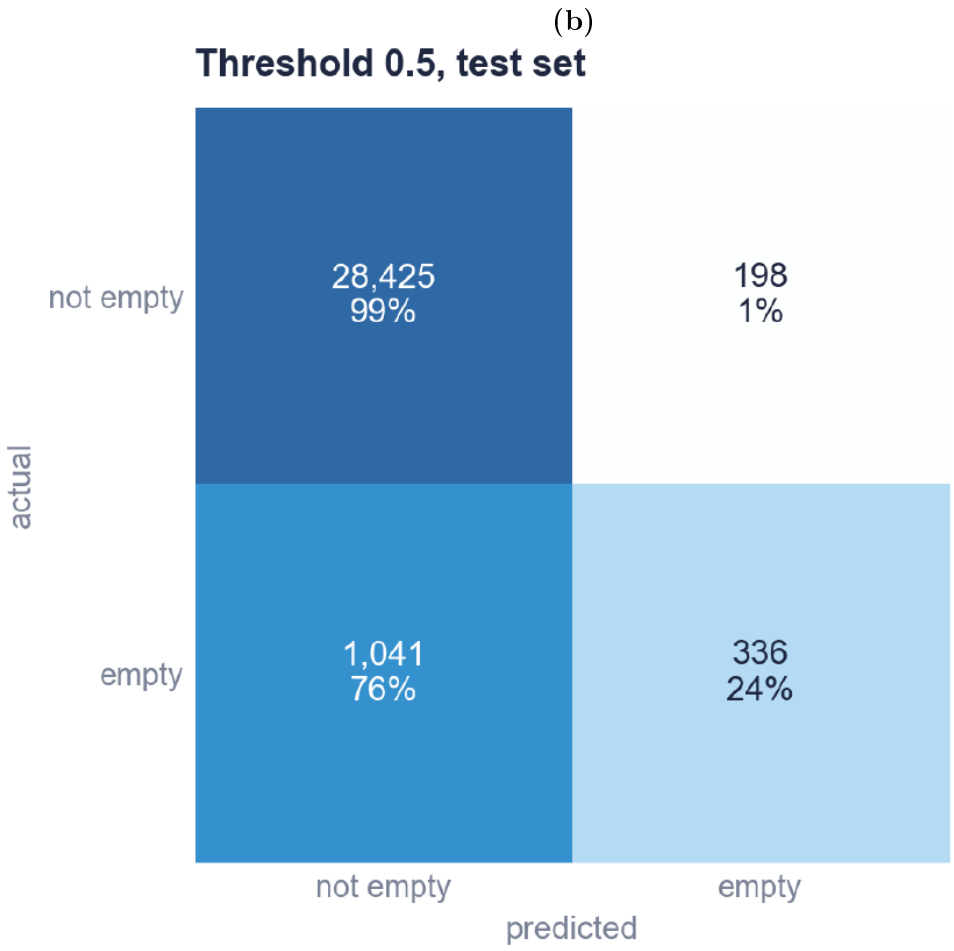


Figure 22: Panels continued.

3.5.6 What drives the prediction

Two views agree (Figure 23). In gain importance, the empty share this evening (30%) and the empty share this morning (17%) are the two most important variables, followed by arrivals today (9%) and the work-station type (8%). In an ablation, where we remove a group of variables and retrain, losing *the station's state today* costs 0.145 AUC (down to 0.762), today's rides 0.026, the calendar 0.015, the station's capacity and type 0.008 and the weather 0.000.

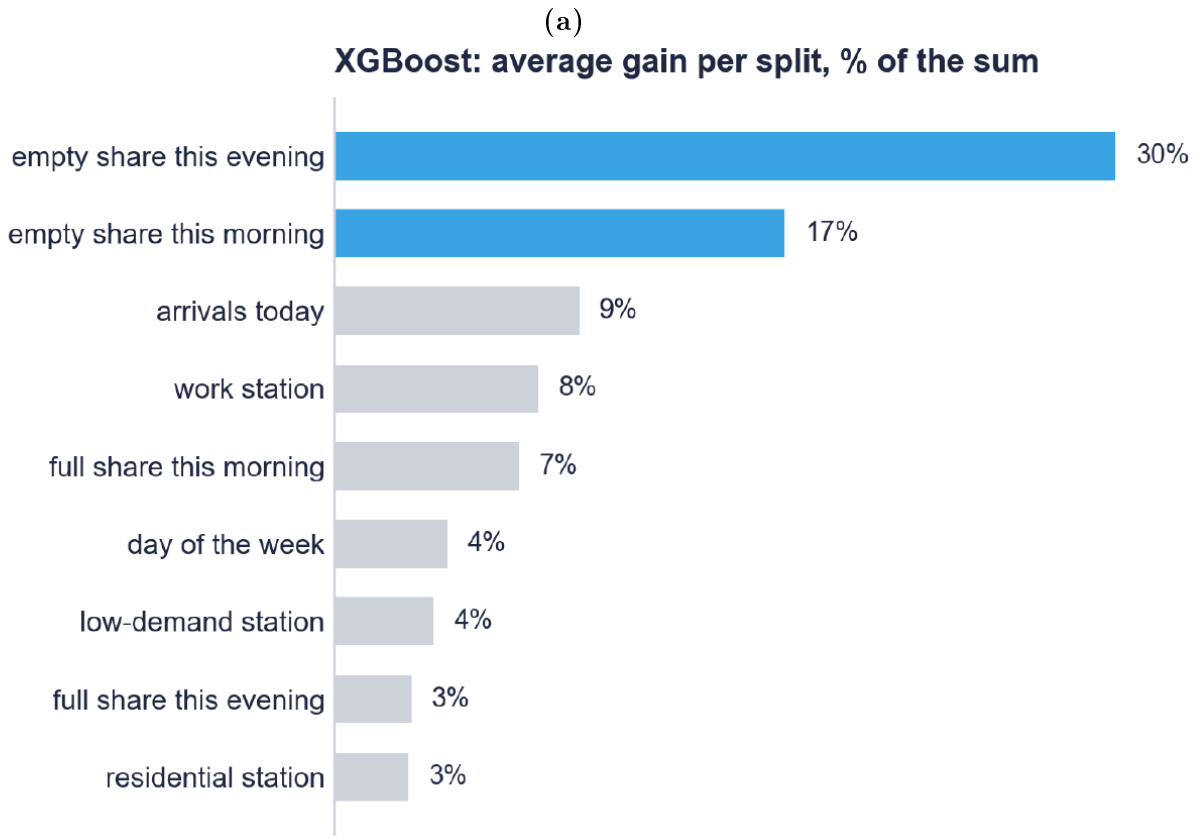


Figure 23: (a) XGBoost importance (average gain per split, top 9 variables). (b) test AUC lost when a group of variables is removed and the model retrained (full model 0.906).

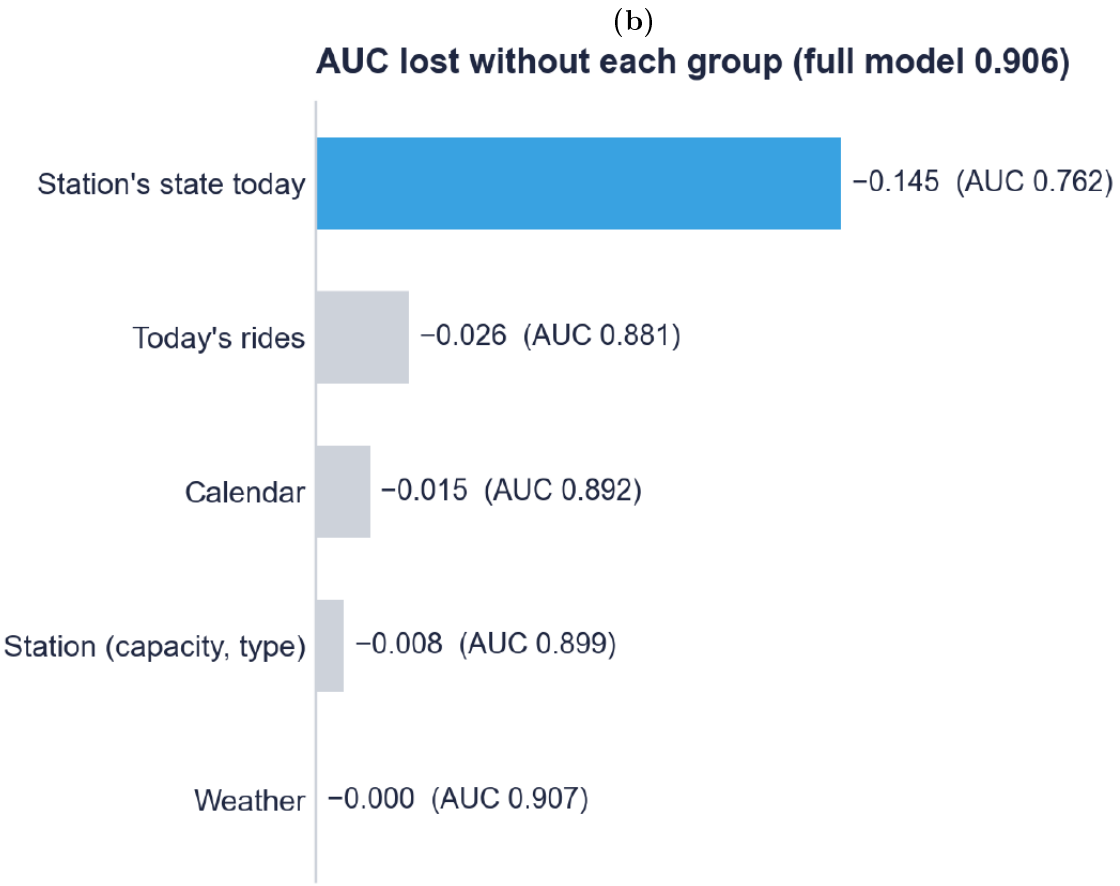


Figure 23: Panels continued.

What this means for Citi Bike

The model does not need weather forecasts, external data or customer information: the station's own state today is enough. A pilot can start with the data Citi Bike already collects, and the running cost of the system is mostly one script each evening. The ablation shows that variables from outside the station's own record add nothing.

3.5.7 From scores to two lists

The same scores yield two operational lists each night (Figure 24).

- **Drop-off list:** the 10% of stations at highest risk. Residential stations are 28% of station-days but **43%** of this list; low-demand stations make up another 33%, work stations only 6%. The next morning 30.6% of these stations are empty for at least an hour, against 4.9% for all stations.
- **Pick-up list:** stations full this evening (full for at least one third of 4–8pm) that the model ranks at low risk, taking the half with the lowest predicted risk, about 67 stations a night. The next morning, **0.1%** of them are empty and **52.5%** are full for at least an hour again. Among *all* stations full this evening, 1.0% are empty and 39% are full the next morning.

Pick-up stations are more often busy mixed (39%) and low-demand (31%) stations than residential (18%) or work (12%) stations. Before collecting, the driver checks the live availability and leaves a reserve.

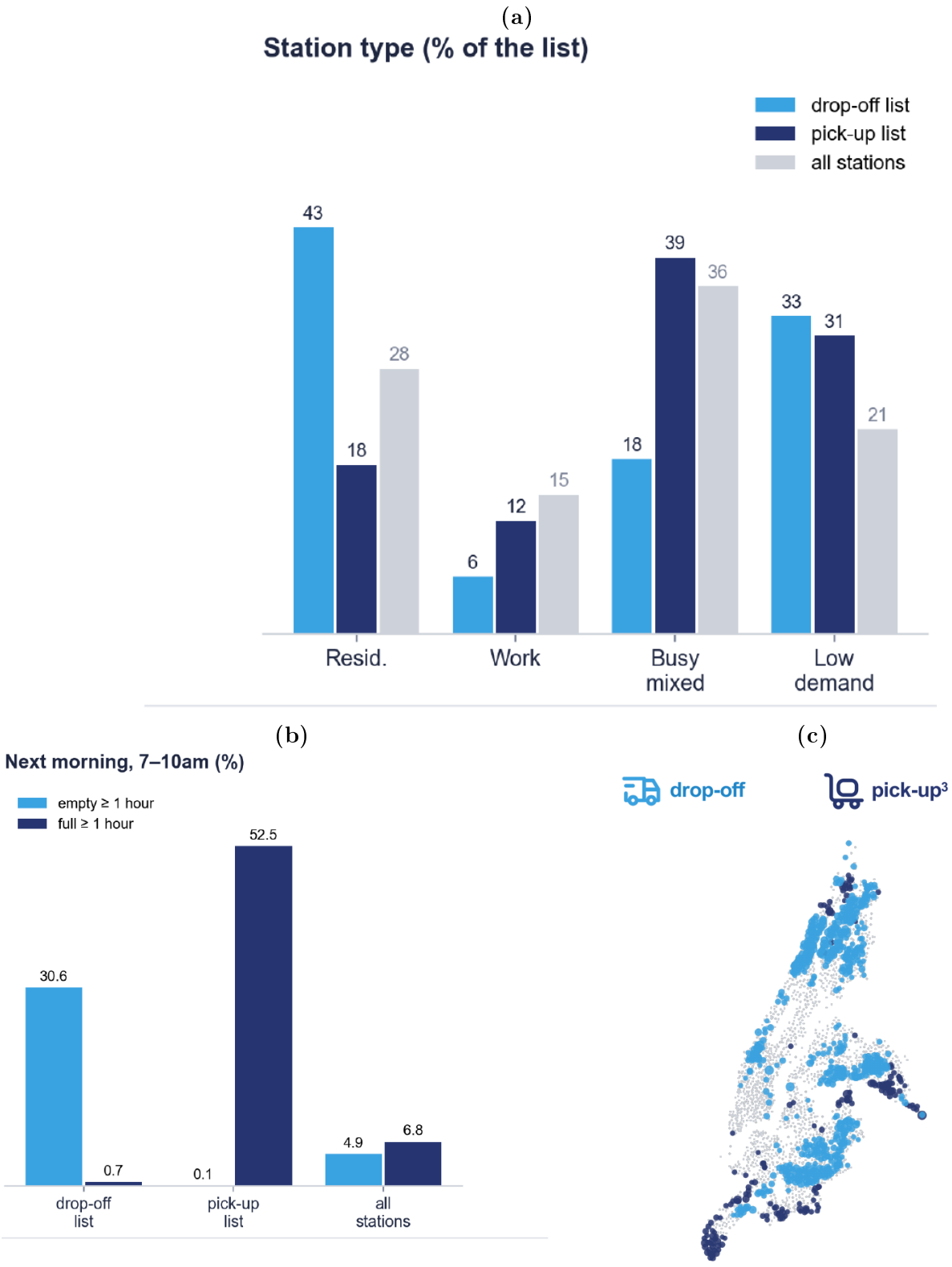


Figure 24: (a) Station type, as a share of each list and of all stations. (b) Outcome the next morning: share of stations empty or full for at least an hour, for the drop-off list, the pick-up list and all stations. (c) Map of stations on the drop-off list (blue) or the pick-up list (navy) on at least 15% of the test nights.

What this means for Citi Bike

The two lists make the trucks' work more efficient in both directions. A truck that refills risky stations and collects from safe, full ones carries load on both legs, whereas a truck that only drops off returns empty. The answer to our opening question is clear: *commuter (residential) stations come first*, refilled with bikes taken from stations that are full tonight and at low risk.

The last analytical step tests whether the model keeps its accuracy over time.

3.6 Robustness: do the models still work months later?

A model fitted in one season may fail in another. We train both the shortage model and the demand regression on **August–October 2025 only** and test them month by month on the following year, with the same settings as before (time-based split on all station-days).

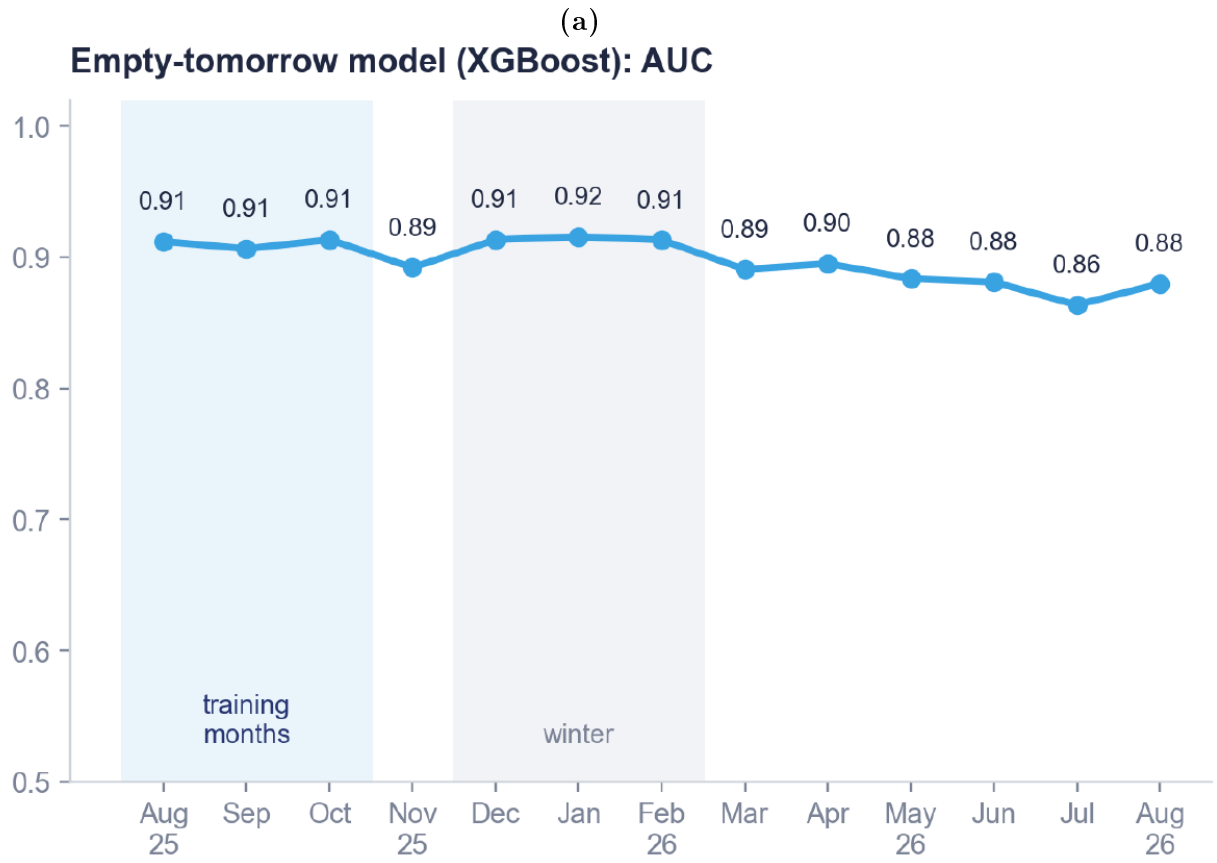


Figure 25: Out-of-time performance of models trained on August–October 2025 (training months shaded in light blue, winter in grey). (a) AUC of the shortage model. (b) R^2 of the demand regression.

(b)

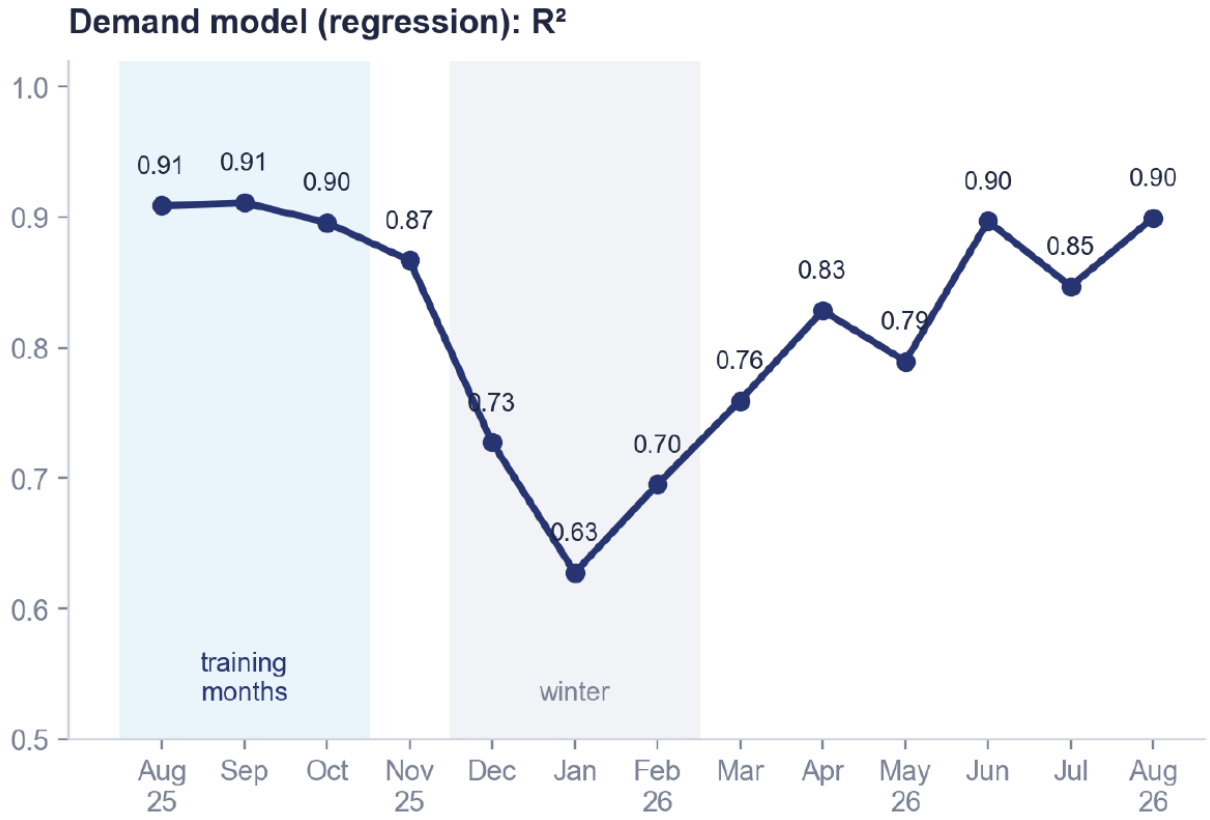


Figure 25: Panels continued.

The shortage model keeps an AUC between 0.86 and 0.92 in every month: 0.911 over the training months, 0.914 in winter (December–February) and 0.877 in June–August 2026. The demand regression falls from an R^2 of 0.905 over the training months to 0.686 in winter, with a low of 0.63 in January, and recovers to 0.881 in summer. The reason is intuitive: volume depends on the season and the weather, so a model that learned autumn demand does not know winter, whereas the *ranking* of stations by risk depends on persistent commuting roles that do not change with the season.

What this means for Citi Bike

The practical conclusion is a maintenance policy. The shortage model can be deployed for a full year without adjustment, and the slight decline one year later (AUC 0.877 in summer 2026) justifies a monthly retraining, which is cheap. A demand forecast for capacity planning, on the other hand, must be retrained for each season or include seasonal terms.

4 What Citi Bike should do

The analysis answers the opening question. This section turns it into actions, gives a rule for how long the nightly list should be, and looks at the customer side.

4.1 Recommendations

1. **Predict.** Run the model every evening and rank all stations by their probability of being empty tomorrow. Plan the route from the top of the list.
2. **Drop off.** Refill residential stations first, before 7am, then the other high-risk stations, including low-demand ones.
3. **Collect.** Take bikes from stations that are full this evening and ranked at low risk, after checking live availability and leaving a reserve.
4. **Fix.** Where a station is empty almost every day, a daily truck is the wrong tool. Add docks or bikes, or a peak-hour valet. This matters because the 10% of stations most often empty account for 46% of empty time on weekday mornings.
5. **Anticipate.** Use weather forecasts to size the fleet of trucks for the city's volume, with extra capacity on Mondays and in winter.
6. **Convert.** Offer memberships to casual riders who repeatedly ride at commuting peaks.

4.2 A simple decision rule for the list size

The length of the nightly list depends on two quantities. Let c be the cost of one stop (driver time, fuel, bikes moved), and v the value of preventing an empty morning at a station (the lost rides, mostly commuters, plus customer goodwill). A stop is worth making if its expected benefit exceeds its cost:

$$h(\alpha)\, v \;\geq\; c \quad\Longleftrightarrow\quad h(\alpha) \;\geq\; \frac{c}{v} = h^{*}, \tag{6}$$

where $h(\alpha)$ is the hit rate of a list of share α of the stations. Since h falls as the list grows (45%, 31%, 19% and 14% for $\alpha = 5, 10, 20, 30\%$), the optimal list is the largest one whose hit rate stays above the break-even h^{*}. For example, an operator for whom $h^{*} = 20\%$ would stop at about the 20% list. A second comparison is the gain relative to the benchmark rule: at equal list size, the model prevents $(0.62 - 0.51) \times S$ additional shortages per night, where S is the number of nightly shortages (about 100 here), i.e. roughly 11 a night. The value of the system is this number multiplied by v and by the number of nights.

Caveat

We have no values for c or v, so the break-even rule gives no estimate of savings. The operator can calibrate it with its own cost data.

4.3 Customer perspective

Because empty and full docks are the main topic of about one review in five, a measurable fall in morning shortages is likely to show up in customer feedback. Before a roll-out, this can be measured: the share of reviews about docks and complaint rates per station, before and after the pilot. The membership offer can be tested in the same way, with a control group of casual riders who are not contacted.

5 Discussion

This section returns to the five hypotheses with the evidence. It then states what our data cannot tell and proposes next steps.

5.1 The hypotheses revisited

	Hypothesis	Result	Evidence
H1	Time explains a station's demand better than the station	Not at station level	Station-day R^2: calendar 0.12, station 0.44, station with last week's rides 0.83. For the whole city, weather explains 78% of daily rides.
H2	Trip patterns separate members and casual riders	Partly	Best AUC 0.72 (random forest), KNN 0.68 at its best k. 28% of casual rides are at commuting peaks, 40% of member rides.
H3	Stations fall into a few behavioural types	Supported	Four types with distinct profiles and opposite daily cycles. Low silhouette (0.11 at $k = 4$): four chosen because each can be acted on.
H4	Weather is a secondary but real driver	Partly	78% of the city's daily rides, +3.5% per °C; removing weather leaves the shortage model's AUC essentially unchanged.
H5	Models lose accuracy in another season	For demand only	Demand R^2 falls to 0.63 in January; the shortage model's AUC stays around 0.91 in winter.

Table 8: The five hypotheses against the results.

Two of the five hypotheses were not confirmed as stated. H1 and H4 show that the usual intuition ("it is the weather and the calendar") is right for the city and wrong for the station. H5 shows that robustness depends on what is predicted: a ranking of risk is more stable than a level of demand.

5.2 Limits

No truck data.
We cannot compare our lists with Citi Bike's actual dispatch, and today's station status already includes today's rebalancing. Our benchmark is therefore a *simple rule*, not what the operator does today, and our gain over the real practice is unknown.

Observed weather, not forecast.
We use observed weather in place of tomorrow's forecast. The ablation shows that weather adds nothing, so this should not change the results.

Daily, not hourly.
"Empty for one hour of 7–10am" is a threshold we chose, and the model plans the night, not daytime rounds.

Clustering is a choice.
The silhouette is low, so the station types are a convenient partition and not natural groups. The type is derived from profiles that use the whole year, which is a possible source of mild information leakage in the date-based test. We expect it to be small since the types are built from hourly *shares* and not from the shortage target, but we did not test it.

Small text sample.
The 500 reviews are mostly negative and 362 of them give one star.

Sampling.
Models are trained on samples (100,000 and 150,000 station-days, 20,000 rides for KNN). Stratified sampling and comparisons with the full table limit the risk, but exact figures would differ slightly with another sample.

5.3 Next steps

- **Add the rebalancing data** to measure what the trucks already do and the gain of our lists over it.
- **Predict hour by hour** to plan daytime truck rounds too.
- **Pilot in one borough**, comparing shortages with and without the lists for a few weeks, which turns the analysis into a causal test.
- **Retrain every month**, since the AUC drops slightly a year later as the network changes.

6 Conclusion

A model that uses only what is known the evening before ranks tomorrow's empty stations well (AUC 0.91), also in winter. With 10% of stations it finds 62% of them, against 51% for the simple rule. The answer to our opening question is to send tonight's trucks first to residential stations, with bikes taken from stations that are full and at low risk. The same data also tell Citi Bike whom to approach for a membership: the casual riders who ride when members ride. The analysis is reproducible and states the limits of the data. A pilot in one borough would turn the prediction into a measured operational result.